

Q1: Two equal capacitors are first connected in series and then in parallel. The ratio of the equivalent capacities in the two cases will be

- (A) 1 : 4
- (B) 1 : 2
- (C) 4 : 1
- (D) 2 : 1

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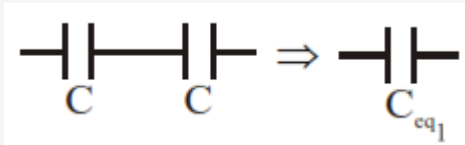
(B) 1 : 2

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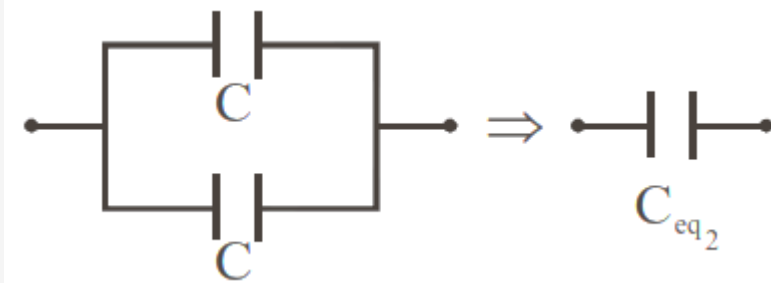
Solution:

For series combination



$$\frac{1}{C_{eq1}} = \frac{1}{C} + \frac{1}{C} \Rightarrow C_{eq1} = \frac{C}{2}$$

For parallel combination



$$C_{eq2} = C + C \Rightarrow C_{eq2} = 2C$$

$$\Rightarrow \frac{C_{eq1}}{C_{eq2}} = \frac{(C/2)}{2C} = \frac{1}{4} = 1 : 4$$

Q2: Each side of a box made of metal sheet in cubic shape is 'a' at room temperature 'T', the coefficient of linear expansion of the metal sheet is ' α '. The metal sheet is heated uniformly, by a small temperature ΔT , so that its new temperature is $T + \Delta T$. Calculate the increase in the volume of the metal box.

- (A) $3a^3\alpha\Delta T$
- (B) $4a^3\alpha\Delta T$
- (C) $\frac{4}{3}\pi a^3\alpha\Delta T$
- (D) $4\pi a^3\alpha\Delta T$

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- (B) $4a^3\alpha\Delta T$
- (C) $4/3\pi a^3\alpha\Delta T$
- (D) $4\pi a^3\alpha\Delta T$

Solution:

$$\Delta V = V\gamma\Delta T$$

$$\Delta V = 3a^3\alpha\Delta T$$

Q3: If an emitter current is changed by 4 mA, the collector current changes by 3.5 mA. The value of β will be

- (A) 3.5
- (B) 0.5
- (C) 0.875
- (D) 7

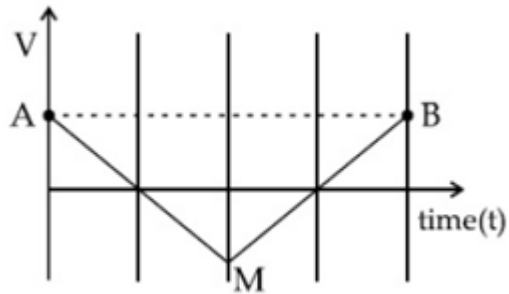
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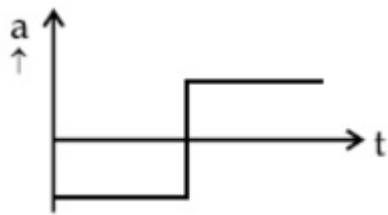
Solution:

$$\begin{aligned}I_e &= I_C + I_B \\ \Rightarrow \Delta I_e &= \Delta I_C + \Delta I_B \\ 4mA &= 3.5mA + \Delta I_B \\ \Rightarrow \Delta I_B &= 0.5mA \\ \Rightarrow \beta &= \frac{\Delta I_C}{\Delta I_B} \\ \beta &= \frac{3.5}{0.5} \\ \beta &= 7\end{aligned}$$

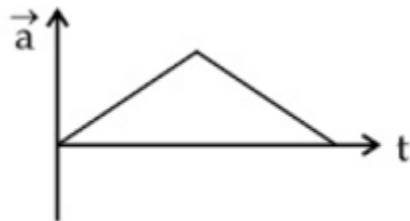
Q4: If the velocity-time graph has the shape AMB, what would be the shape of the corresponding acceleration-time graph?



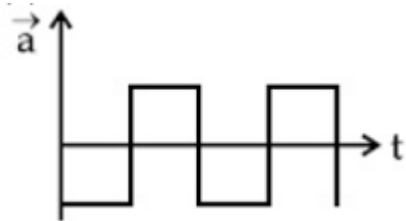
(A)



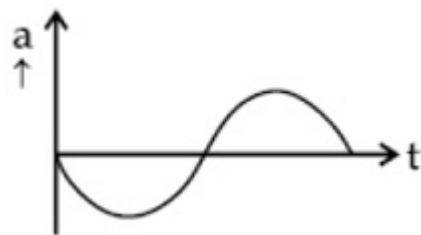
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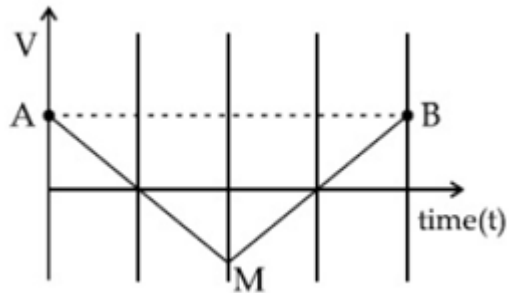
(C)



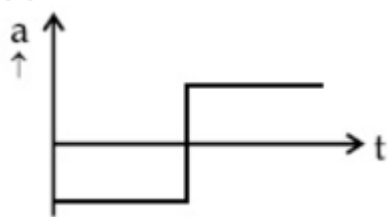
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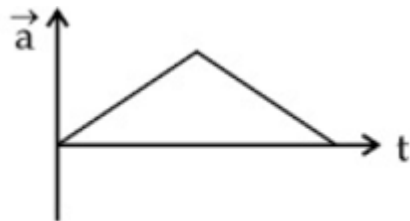
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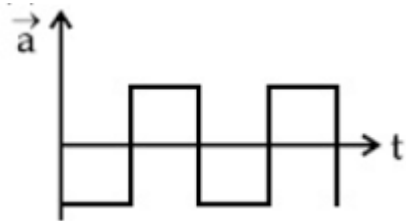
(A)



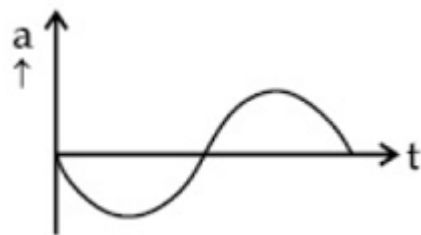
(B)



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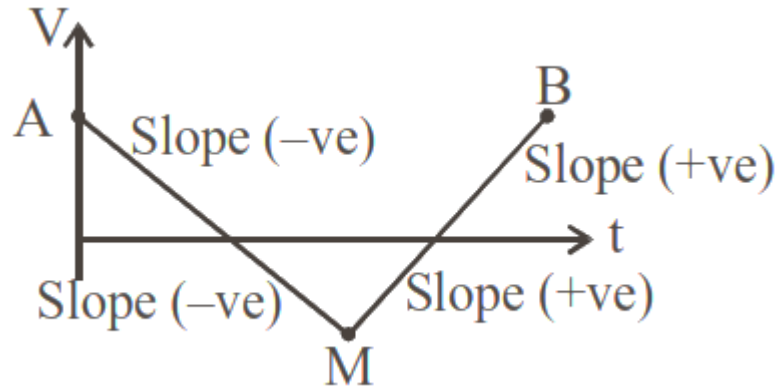


(D)

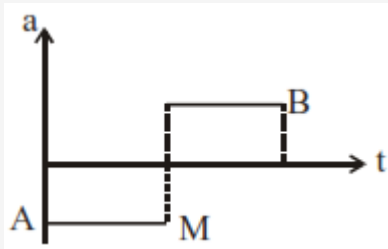


Solution:

Slope of $v - t$ graph gives acceleration



$\Rightarrow$ Acceleration will be



Q5: The work done by a gas molecule in an isolated system is given by, $W = \alpha \beta^2 e^{-\frac{x^2}{\alpha k T}}$, where x is the displacement, k is the Boltzmann constant and T is the temperature. α and β are constants. Then the dimensions of β will be

- (A) $[M^0 L T^0]$
- (B) $[M L T^{-2}]$
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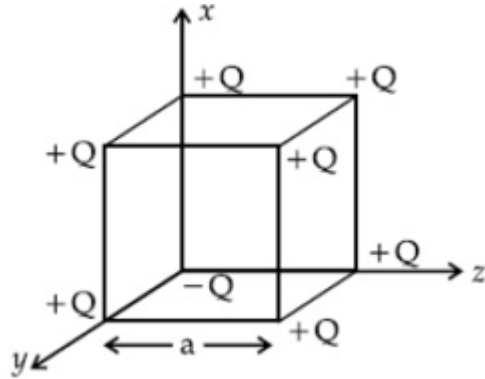
$$\frac{x^2}{\alpha kT} \rightarrow \text{dimensionless}$$

$$\Rightarrow [\alpha] = \frac{[x^2]}{[kT]} = \frac{L^2}{ML^2T^{-2}} = M^{-1}T^2$$

$$\text{Now } [W] = [\alpha] [\beta]^2$$

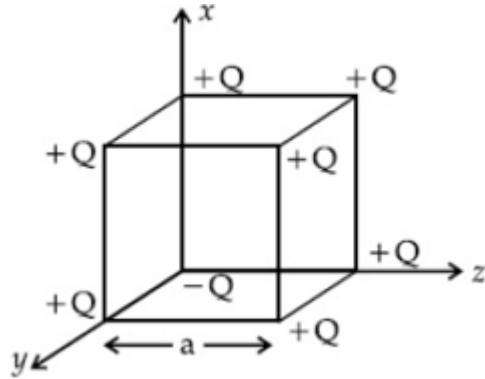
$$[\beta] = \sqrt{\frac{ML^2T^{-2}}{M^{-1}T^2}} = M^1L^1T^{-2}$$

Q6: A cube of side 'a' has point charges +Q located at each of its vertices except at the origin where the charge is -Q. The electric field at the centre of cube is:



- (A) $\frac{-Q}{3\sqrt{3}\pi\epsilon_0 a^2} (\hat{x} + \hat{y} + \hat{z})$
- (B) $\frac{-2Q}{3\sqrt{3}\pi\epsilon_0 a^2} (\hat{x} + \hat{y} + \hat{z})$
- (C) $\frac{2Q}{3\sqrt{3}\pi\epsilon_0 a^2} (\hat{x} + \hat{y} + \hat{z})$
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(D) $\frac{Q}{3\sqrt{3}\pi\epsilon_0 a^2} (\hat{x} + \hat{y} + \hat{z})$

Solution:

We can replace $-Q$ charge at origin by $+Q$ and $-2Q$. Now due to $+Q$ charge at every corner of cube. Electric field at center of cube is zero so now net electric field at only due to $-2Q$ charge at origin

$$\vec{E} = \frac{kq\vec{r}}{r^3} = \frac{1(-2Q)\frac{a}{2}(\hat{x}+\hat{y}+\hat{z})}{4\pi\epsilon_0\left(\frac{a}{2}\sqrt{3}\right)^3}$$

$$\vec{E} = \frac{-2Q(\hat{x}+\hat{y}+\hat{z})}{3\sqrt{3}\pi a^2\epsilon_0}$$

Q7: In a Young's double slit experiment, the width of the one of the slit is three times the other slit. The amplitude of the light coming from a slit is proportional to the slit-width. Find the ratio of the maximum to the minimum intensity in the interference pattern.

(A) 4 : 1

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Solution:

Amplitude $\propto$ Width of slit

$$\Rightarrow A_2 = 3A_1$$

$$\frac{I_{\max}}{I_{\min}} = \left(\frac{\sqrt{I_1} + \sqrt{I_2}}{\sqrt{I_1} - \sqrt{I_2}} \right)^2$$

$\therefore$ Intensity $I \propto A^2$

$$\Rightarrow \frac{I_{\max}}{I_{\min}} = \left(\frac{A_1 + 3A_1}{|A_1 - 3A_1|} \right)^2$$

$$= \left(\frac{4A_1}{2A_1} \right)^2 = 4 : 1$$

Q8: If Y , K and η are the values of Young's modulus, bulk modulus and modulus of rigidity of any material respectively. Choose the correct relation for these parameters.

(A) $Y = \frac{9K\eta}{3K-\eta} - N/m^2$

(B) $Y = \frac{9K\eta}{2\eta-3K} - N/m^2$

(C) $K = \frac{Y\eta}{9\eta-3Y} - N/m^2$

(D) $\eta = \frac{3YK}{9K-Y} - N/m^2$

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Solution:

Y – Young modulus, K – Bulk modulus,

η - modulus of rigidity

We know that

$$y = 3k(1 - 2\sigma)$$

$$\sigma = \frac{1}{2} \left(1 - \frac{y}{3k} \right) \dots\dots(i)$$

$$y = 2\eta(1 + \sigma)$$

$$\sigma = \frac{y}{2\eta} - 1 \dots\dots(ii)$$

From Eq. (i) and Eq. (ii)

$$\frac{1}{2} \left(1 - \frac{Y}{3k} \right) = \frac{y}{2\eta} - 1$$

$$1 - \frac{y}{3k} = \frac{y}{\eta} - 2$$

$$\frac{y}{3k} = 3 - \frac{y}{\eta}$$

$$\frac{y}{3k} = \frac{3\eta - y}{\eta}$$

$$\frac{\eta y}{3k} = 3\eta - y$$

$$k = \frac{\eta y}{9\eta - 3y}$$

Q9: Four identical particles of equal masses 1 kg move to move along the circumference of a circle of radius 1 m under the action of their own mutual gravitational attraction. The speed of each particle will be

(A) $\sqrt{\frac{G}{2}(2\sqrt{2} - 1)}$

(B) $\sqrt{G(1 + 2\sqrt{2})}$

(C) $\frac{\sqrt{(1+2\sqrt{2})G}}{2}$

(D) $\sqrt{\frac{G}{2}(1 + 2\sqrt{2})}$

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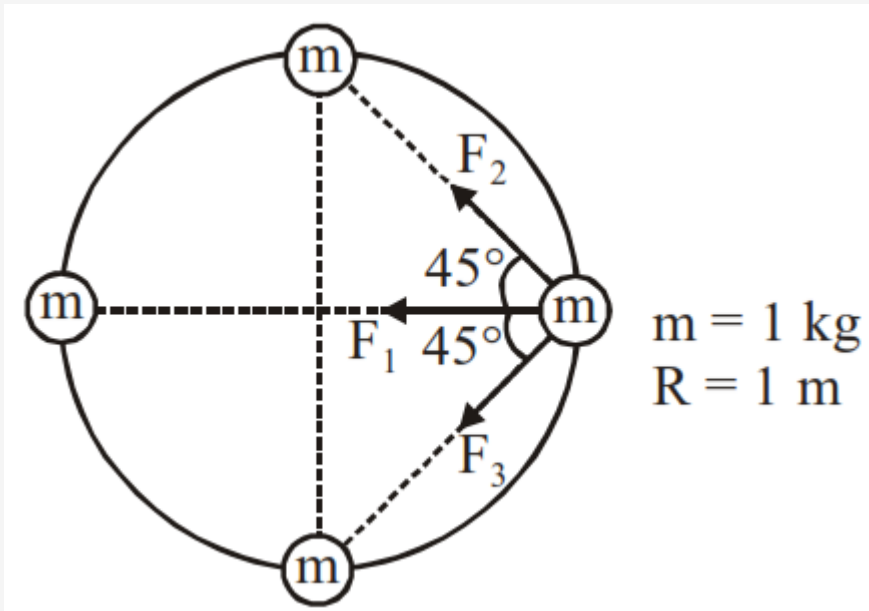
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Solution:



$$F_1 = \frac{Gmm}{(2R)^2} = \frac{Gm^2}{4R^2}$$

$$F_2 = \frac{Gmm}{(\sqrt{2}R)^2} = \frac{Gm^2}{2R^2}$$

$$F_3 = \frac{Gmm}{(\sqrt{2}R)^2} = \frac{Gm^2}{2R^2}$$

$$\Rightarrow F_{net} = F_1 + F_2 \cos 45^\circ + F_3 \cos 45^\circ$$

$$= \frac{Gm^2}{4R^2} + \frac{Gm^2}{2R^2} \frac{1}{\sqrt{2}} + \frac{Gm^2}{2R^2} \frac{1}{\sqrt{2}}$$

$$= \frac{Gm^2}{R^2} \left(\frac{1}{4} + \frac{1}{2\sqrt{2}} + \frac{1}{2\sqrt{2}} \right)$$

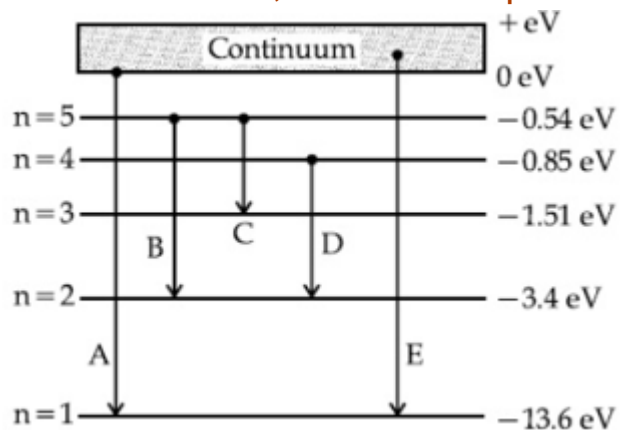
$$= \frac{Gm^2}{R^2} \left(\frac{1}{4} + \frac{1}{\sqrt{2}} \right) = \frac{Gm^2}{4R^2} (1 + 2\sqrt{2})$$

$$F_{net} = \frac{Gm^2}{4R^2} (1 + 2\sqrt{2}) = \frac{mv^2}{R}$$

$$\Rightarrow v = \frac{\sqrt{G(1+2\sqrt{2})}}{2}$$

Q10: In the given figure, the energy levels of hydrogen atom have been shown along with some transitions marked A, B, C, D and E.

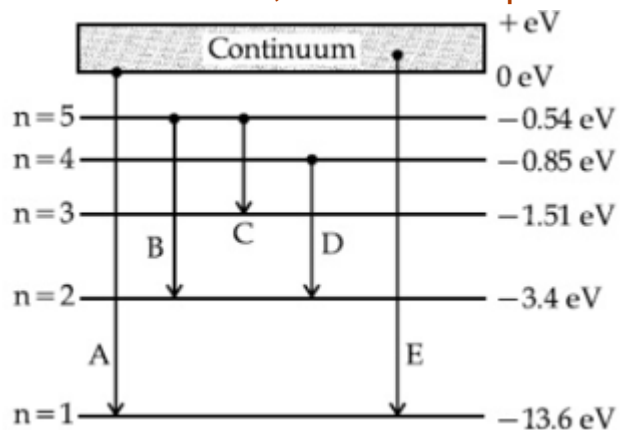
The transitions A, B and C respectively represent:



- (A) The series limit of Lyman series, third member of Balmer series and second member of Paschen series
- (B) The ionization potential of hydrogen, second member of Balmer series and third member of Paschen series
- (C) The first member of the Lyman series, third member of Balmer series and second member of Paschen series
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- (C) The first member of the Lyman series, third member of Balmer series and second member of Paschen series
- (D) The series limit of Lyman series, second member of Balmer series and second member of Paschen series

Solution:

A $\rightarrow$ Series limit of Lyman series

B $\rightarrow$ Third member of Balmer series

C $\rightarrow$ Second member of Paschen series.

Q11: A current through a wire depends on time as $i = \alpha_0 t + \beta t^2$ where $\alpha_0 = 20\text{A/s}$ and $\beta = 8\text{ As}^{-2}$. Find the charge crossed through a section of the wire in 15s.

- (A) 11250 C
- (B) 2250 C
- (C) 260 C
- (D) 2100 C

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Solution:

$$i = 20t + 8t^2$$

$$i = \frac{dq}{dt} \Rightarrow \int dq = \int i dt$$

$$\Rightarrow q = \int_0^{15} (20t + 8t^2) dt$$

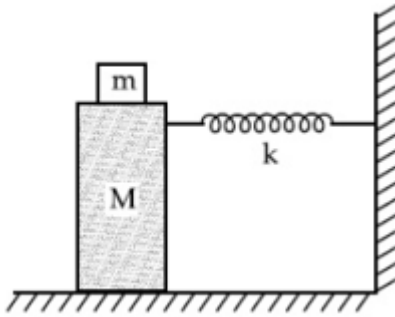
$$q = \left(\frac{20t^2}{2} + \frac{8t^3}{3} \right)_0^{15}$$

$$q = 10 \times (15)^2 + \frac{8(15)^3}{3}$$

$$q = 2250 + 9000$$

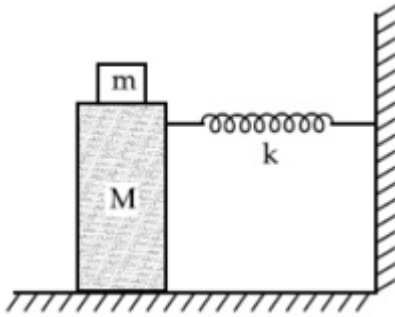
$$q = 11250\text{C}$$

Q12: In the given figure, a mass M is attached to a horizontal spring which is fixed on one side to a rigid support. The spring constant of the spring is k . The mass oscillates on a frictionless surface with time period T and amplitude A . When the mass is in equilibrium position, as shown in the figure, another mass m is gently fixed upon it. The new amplitude of oscillation will be:



- (A) $A\sqrt{\frac{M-m}{M}}$
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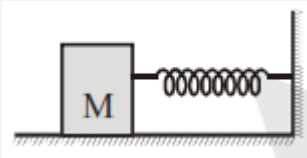
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Solution:



Momentum of system remains conserved

$$P_i = P_f$$

$$MA\omega = (m + M) A' \omega'$$

$$MA\sqrt{\frac{k}{M}} = (m + M) A' \sqrt{\frac{k}{m+M}}$$

$$A' = A\sqrt{\frac{M}{M+m}}$$

Q13: Match List-I with List-II

	List I		List II
A	Isothermal	(i)	Pressure constant
B	Isochoric	(ii)	Temperature constant
C	Adiabatic	(iii)	Volume constant
D	Isobaric	(iv)	Heat content is constant

Choose the correct answer from the options given below

- (A) A – (iii), B – (ii), C – (i), D – (iv)
- (B) A – (ii), B – (iv), C – (iii), D – (i)
- (C) A – (i), B – (iii), C – (ii), D – (iv)
- (D) A – (ii), B – (iii), C – (iv), D – (i)

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(C) A – (i), B – (iii), C – (ii), D – (iv)

(D) A – (ii), B – (iii), C – (iv), D – (i)

Solution:

(a) Isothermal $\Rightarrow$ Temperature constant

(a) $\rightarrow$ (ii)

(b) Isochoric $\Rightarrow$ Volume constant

(a) $\rightarrow$ (iii)

(c) Adiabatic $\Rightarrow$ Change in heat = 0 $\Rightarrow$ Heat content is constant (*c*) $\rightarrow$ (*iv*)

(a) $\rightarrow$ (iv)

(d) Isobaric $\Rightarrow$ Pressure constant

(d) $\rightarrow$ (i)

Q14: The focal length f is related to the radius of curvature r of the spherical convex mirror by

(A) $f = -r$

(B) $f = -\frac{1}{2}r$

(C) $f = r$

(D) $f = +\frac{1}{2}r$

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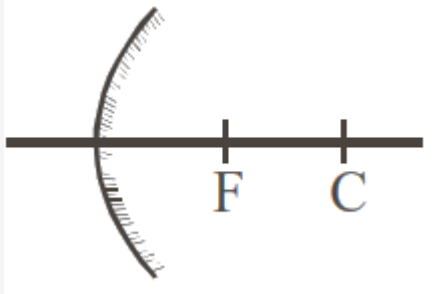
(B) $f = -\frac{1}{2}r$

(C) $f = r$

(D) $f = +\frac{1}{2}r$

Solution:

For convex mirror, focus is behind the mirror



$$\Rightarrow f = +\frac{r}{2}$$

Q15: Two stars of masses m and $2m$ at a distance d rotate about their common centre of mass in free space. The period of revolution is

(A) $2\pi\sqrt{\frac{3Gm}{d^3}}$

(B) $2\pi\sqrt{\frac{d^3}{3Gm}}$

(C) $\frac{1}{2\pi}\sqrt{\frac{d^3}{3Gm}}$

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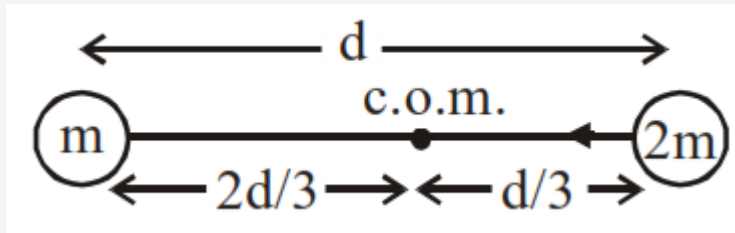
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(D) $\frac{1}{2\pi}\sqrt{\frac{3Gm}{d^3}}$

Solution:



$$F = \frac{G(2m)m}{d^2} = (2m)\omega^2 \left(\frac{d}{3}\right)$$

$$\frac{Gm}{d^2} = \omega^2 \frac{d}{3}$$

$$\Rightarrow \omega^2 = \frac{3Gm}{d^3}$$

$$\Rightarrow \omega = \sqrt{\frac{3Gm}{d^3}}$$

$$\Rightarrow T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{d^3}{3Gm}}$$

Q16: Moment of inertia (M.I.) of four bodies, having same mass and radius, are reported as;

I_1 = M.I. of thin circular ring about its diameter,

I_2 = M.I. of circular disc about an axis perpendicular to disc and going through the centre,

I_3 = M.I. of solid cylinder about its axis and

I_4 = M.I. of solid sphere about its diameter.

Then:

(A) $I_1 + I_3 < I_2 + I_4$

(B) $I_1 I_2 = I_3 + \frac{5}{2} I_4$

(C) $I_1 + I_2 = I_3 < I_4$

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(D) $I_1 = I_2 = I_3 > I_4$

Solution:

Ring $I_1 = \frac{MR^2}{2}$ about diameter

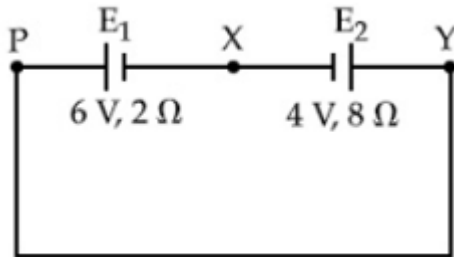
Disc $I_2 = \frac{MR^2}{2}$

Solid cylinder $I_3 = \frac{MR^2}{2}$

Solid sphere $I_4 = \frac{2}{5} MR^2$

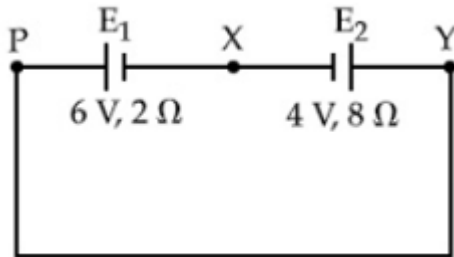
$I_1 = I_2 = I_3 > I_4$

Q17: A cell E_1 of emf 6 V and internal resistance 2Ω is connected with another cell E_2 of emf 4 V and internal resistance 8Ω (as shown in the figure). The potential difference across points X and Y is:



- (A) 10.0 V
- (B) 5.6 V
- (C) 3.6 V
- (D) 2.0 V

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- (A) 10.0 V
- (B) 5.6 V
- (C) 3.6 V
- (D) 2.0 V

Solution:

$$I = \frac{6-4}{10} = \frac{1}{5} A$$

$$V_x + 4 + 8 \times \frac{1}{5} - V_y = 0$$

$$V_x - V_y = -5.6 \Rightarrow |V_x - V_y| = 5.6 V$$

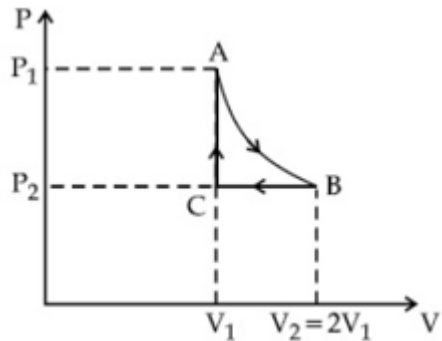
Q18: n mole of a perfect gas undergoes a cyclic process ABCA (see figure) consisting of the following processes.

$A \rightarrow B$: Isothermal expansion at temperature T so that the volume is doubled from V_1 to $V_2 = 2V_1$ and pressure changes from P_1 to P_2 .

$B \rightarrow C$: Isobaric compression at pressure P_2 to initial volume V_1 .

$C \rightarrow A$: Isochoric change leading to change of pressure from P_2 to P_1 .

Total work done in the complete cycle ABCA is:



- (A) 0
- (B) $nRT \left(\ln 2 - \frac{1}{2} \right)$
- (C) $nRT \ln 2$
- (D) $nRT \left(\ln 2 + \frac{1}{2} \right)$

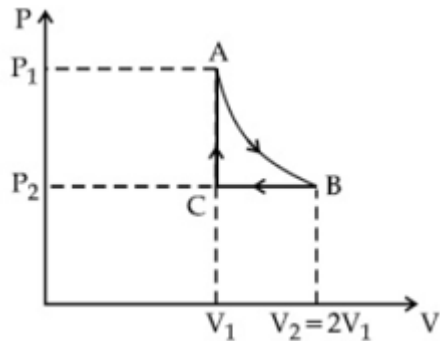
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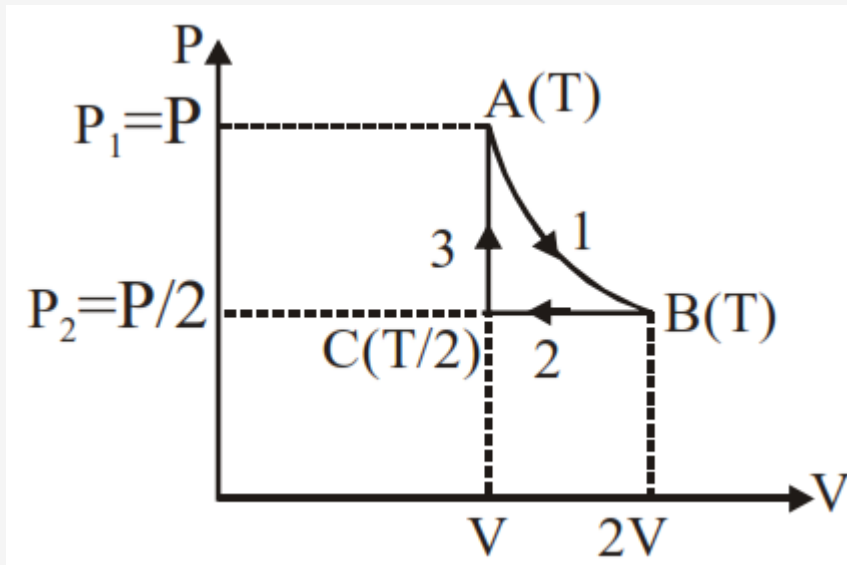
(D) $nRT \left(\ln 2 + \frac{1}{2} \right)$

Solution:

$$W_{\text{Isothermal}} = nRT \ln\left(\frac{v_2}{v_1}\right)$$

$$W_{\text{Isobaric}} = P\Delta V = nR\Delta T$$

$$W_{\text{Isochoric}} = 0$$



$$W_1 = nRT \ln\left(\frac{2V}{V}\right) = nRT \ln 2$$

$$W_2 = \frac{P}{2}(V - 2V) = \frac{-PV}{2} = \frac{-nRT}{2}$$

$$W_3 = 0$$

$$W_{\text{net}} = nRT \left(\ln 2 - \frac{1}{2}\right)$$

Q19: Given below are two statements:

Statement I: Two photons having equal linear momenta have equal wavelength.

Statement II: If the wavelength of photon is decreased, then the momentum and energy of a photon will also decrease.

In the light of the above statements, choose the correct answer from the options given below.

- (A) Statement I is true but Statement II is false
- (B) Both Statement I and Statement II are false
- (C) Statement I is false but Statement II is true
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- (B) Both Statement I and Statement II are false
- (C) Statement I is false but Statement II is true
- (D) Both Statement I and Statement II are true

Solution:

If linear momentum are equal then wavelength also equal

$$p = \frac{h}{\lambda}, E = \frac{hc}{\lambda}$$

On decreasing wavelength, momentum and energy of photon increases.

Q20: Consider two satellites S_1 and S_2 with periods of revolution 1 hr. and 8hr. respectively revolving around a planet in circular orbits. The ratio of angular velocity of satellite S_1 to the angular velocity of satellite S_2 is

- (A) 1 : 8
- (B) 2 : 1
- (C) 1 : 4
- (D) 8 : 1

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(A) 1 : 8

(B) 2 : 1

(C) 1 : 4

(D) 8 : 1

Solution:

$$\frac{T_1}{T_2} = \frac{1}{8}$$

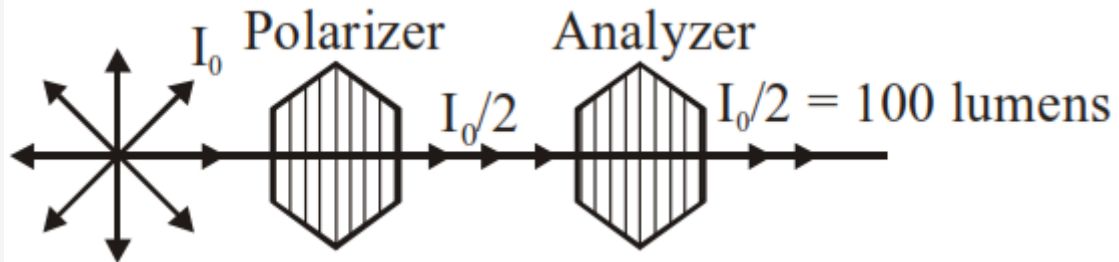
$$\frac{2\pi/\omega_1}{2\pi/\omega_2} = \frac{1}{8}$$

$$\frac{\omega_1}{\omega_2} = \frac{8}{1}$$

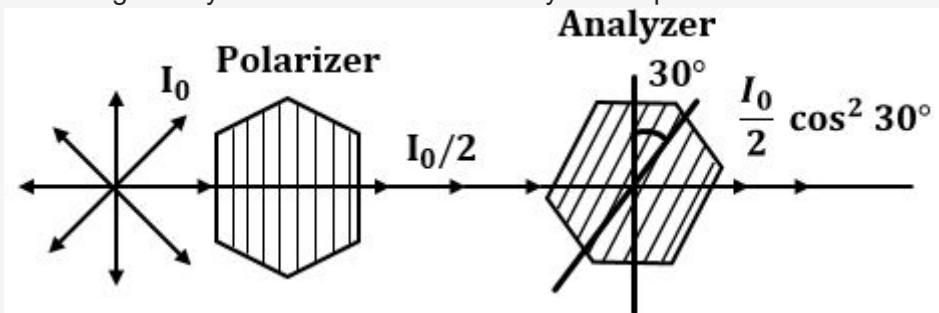
Q21: An unpolarised light beam is incident on the polarizer of a polarization experiment and the intensity of light beam emerging from the analyser is measured as 100 Lumens. Now, if the analyser is rotated around the horizontal axis (direction of light) by 30° in clockwise direction, the intensity of emerging light will be _____ Lumens.

Q21: An unpolarised light beam is incident on the polarizer of a polarization experiment and the intensity of light beam emerging from the analyser is measured as 100 Lumens. Now, if the analyser is rotated around the horizontal axis (direction of light) by 30° in clockwise direction, the intensity of emerging light will be _____ Lumens.

Solution:



Assuming initially axis of Polarizer and Analyzer are parallel



Now emerging intensity $\frac{I_0}{2} \cos^2 30^\circ$

$$= 100 \left(\frac{\sqrt{3}}{2} \right)^2 = 100 \times \frac{3}{4} = 75$$

$\therefore$ Ans: 75

Q22: A resonance circuit having inductance and resistance $2 \times 10^{-4}\text{H}$ and $6.28\ \Omega$ respectively oscillates at 10 MHz frequency. The value of quality factor of this resonator is _____. [$\pi = 3.14$]

Q22: A resonance circuit having inductance and resistance $2 \times 10^{-4} \text{H}$ and 6.28Ω respectively oscillates at 10 MHz frequency. The value of quality factor of this resonator is _____. [$\pi = 3.14$]

2000.00

Solution:

Given: $L = 2 \times 10^{-4} \text{H}$

$R = 6.28 \Omega$

$f = 10 \text{MHz} = 10^7 \text{Hz}$

Since quality factor,

$$Q = \omega_0 \frac{L}{R} = 2\pi f \frac{L}{R}$$

$$\therefore Q = 2\pi \times 10^7 \times \frac{2 \times 10^{-4}}{6.28}$$

$$Q = 2 \times 10^3 = 2000$$

$\therefore$ Ans. is 2000

Q23: A common transistor radio set requires 12 V (D.C) for its operation. The D.C. source is constructed by using a transformer and a rectifier circuit, which are operated at 220 V (A.C.) on standard domestic A.C. supply. The number of turns of secondary coil are 24, then the number of turns of primary are_____.

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440.00

Solution:

$$\frac{N_P}{N_S} = \frac{V_P}{V_S}$$

$$\frac{N_P}{24} = \frac{220}{12}$$

$$N_P = \frac{220 \times 24}{12}$$

$$N_P = 440$$

Ans. 440 turns

Q24: An electromagnetic wave of frequency 5GHz, is travelling in a medium whose relative electric permittivity and relative magnetic permeability both are 2. Its velocity in this medium is _____ $\times 10^7$ m/s.

Q24: An electromagnetic wave of frequency 5GHz, is travelling in a medium whose relative electric permittivity and relative magnetic permeability both are 2. Its velocity in this medium is _____ $\times 10^7$ m/s.

15.00

Solution:

Given: Frequency of wave $f = 5$ GHz

$$= 5 \times 10^9 \text{ Hz}$$

Relative permittivity, $\epsilon_r = 2$

And Relative permeability, $\mu_r = 2$

Since speed of light in a medium is given by

$$v = \frac{1}{\sqrt{\mu\epsilon}} = \frac{1}{\sqrt{\mu_r\mu_0\epsilon_r\epsilon_0}}$$

$$v = \frac{1}{\sqrt{\mu_r\epsilon_r}} \frac{1}{\sqrt{\mu_0\epsilon_0}} = \frac{C}{\sqrt{\mu_r\epsilon_r}}$$

Where C is speed of light in vacuum

$$\therefore v = \frac{3 \times 10^8}{\sqrt{4}} = \frac{30 \times 10^7}{2} \text{ m/s}$$

$$= 15 \times 10^7 \text{ m/s}$$

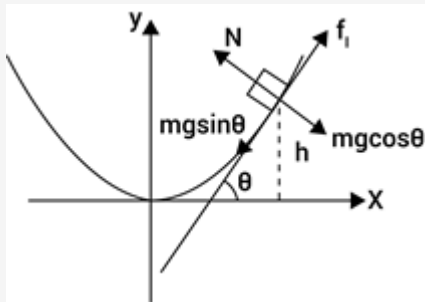
$\therefore$ Ans. is 15

Q25: An inclined plane is bent in such a way that the vertical cross-section is given by $y = \frac{x^2}{4}$ where y is in vertical and x in horizontal direction. If the upper surface of this curved plane is rough with coefficient of friction $\mu = 0.5$, the maximum height in cm at which a stationary block will not slip downward is _____ cm.

Q25: An inclined plane is bent in such a way that the vertical cross-section is given by $y = \frac{x^2}{4}$ where y is in vertical and x in horizontal direction. If the upper surface of this curved plane is rough with coefficient of friction $\mu = 0.5$, the maximum height in cm at which a stationary block will not slip downward is _____ cm.

25.00

Solution:



$$N = mg \cos \theta$$

$$f_l = mg \sin \theta \quad \dots (1)$$

$$f_l = \mu N = \mu mg \cos \theta \quad \dots (2)$$

Using (1) and (2),

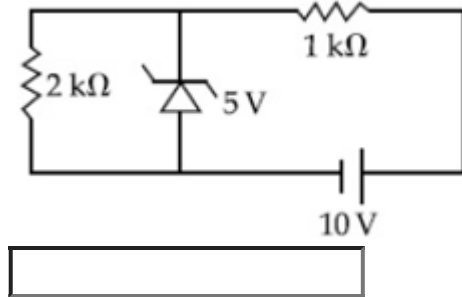
$$\tan \theta = \mu$$

$$\frac{dy}{dx} = \mu$$

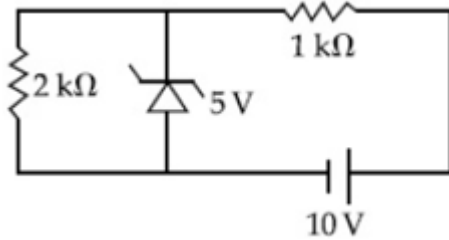
$$\frac{x}{2} = \frac{1}{2}, \quad x = 1$$

$$\text{So, } h = \frac{x^2}{4} = 0.25 \text{ m} = 25 \text{ cm}$$

Q26: In connection with the circuit drawn below, the value of current flowing through $2k\Omega$ resistor is _____ $\times 10^{-4}A$.



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25.00

Solution:

Current through $2k\Omega$ resistance

$$I = \frac{5}{2 \times 10^3} = 2.5 \times 10^{-3} A$$

$$I = 25 \times 10^{-4} A$$

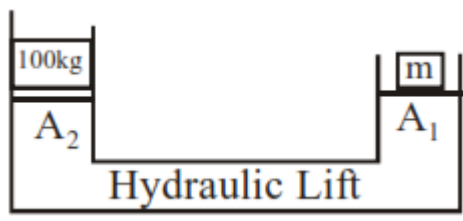
Q27: A hydraulic press can lift 100 kg when a mass 'm' is placed on the smaller piston. It can lift _____ kg when the diameter of the larger piston is increased by 4 times and that of the smaller piston is decreased by 4 times keeping the same mass 'm' on the smaller piston.

Q27: A hydraulic press can lift 100 kg when a mass 'm' is placed on the smaller piston. It can lift _____ kg when the diameter of the larger piston is increased by 4 times and that of the smaller piston is decreased by 4 times keeping the same mass 'm' on the smaller piston.

25600.00

Solution:

Using Pascals law



$$\frac{100 \times g}{A_2} = \frac{mg}{A_1} \dots\dots(1)$$

Let m mass can lift M_0 in second case then

$$\frac{M_0 g}{16A_2} = \frac{mg}{A_1/16} \dots\dots(2)$$

$$\left\{ \text{Since } A = \frac{\pi d^2}{4} \right\}$$

From equation (1) and (2) we get

$$\frac{M_0}{16 \cdot 100} = 16$$

$$\Rightarrow M_0 = 25600kg$$

Q28: The coefficient of static friction between a wooden block of mass 0.5 kg and a vertical rough wall is 0.2. The magnitude of horizontal force should be applied on the block to keep it adhere to the wall will be _____N.

[$g = 10 \text{ ms}^{-2}$]

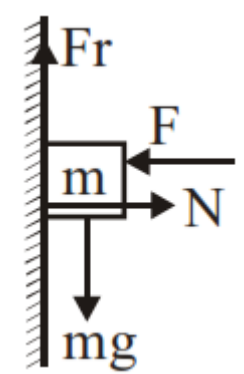
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[$g = 10 \text{ ms}^{-2}$]

25.00

Solution:

F.B.D. of the block is shown in the diagram



Since block is at rest therefore

$$Fr - mg = 0 \dots\dots (1)$$

$$F - N = 0 \dots\dots(2)$$

$$fr \leq \mu N$$

In limiting case

$$fr = \mu N = \mu F \dots\dots(3)$$

Using eq. (1) and (3)

$$\therefore \mu F = mg$$

$$\Rightarrow F = \frac{0.5 \times 10}{0.2} = 25N$$

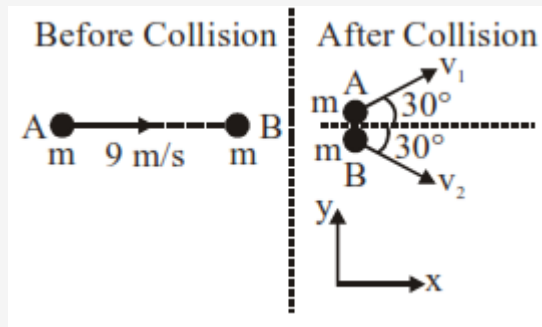
Ans. 25.00

Q29: A ball with a speed of 9 m/s collides with another identical ball at rest. After the collision, the direction of each ball makes an angle of 30° with the original direction. The ratio of velocities of the balls after collision is $x : Y$, where x is _____.

Q29: A ball with a speed of 9 m/s collides with another identical ball at rest. After the collision, the direction of each ball makes an angle of 30° with the original direction. The ratio of velocities of the balls after collision is $x : Y$, where x is _____.

1.00

Solution:



From conservation of momentum along y – axis. $\vec{P}_{iy} = \vec{P}_{fy}$

$$0 + 0 = mv_1 \sin 30^\circ \hat{j} + mv_2 \sin 30^\circ (-\hat{j})$$

$$mv_2 \sin 30^\circ \hat{j} = mv_1 \sin 30^\circ$$

$$v_2 = v_1 \text{ or } \frac{v_1}{v_2} = 1$$

Ans. 1

Q30: An audio signal $v_m = 20 \sin 2\pi(1500t)$ amplitude modulates a carrier $v_c = 80 \sin 2\pi(100,000t)$.

The value of percent modulation is_____.

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The value of percent modulation is_____.

Solution:

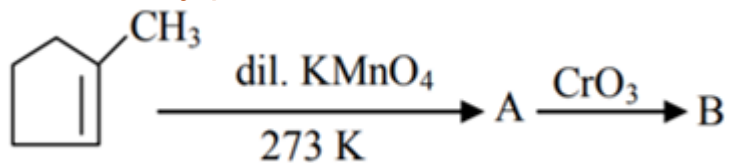
$$\% \text{modulation} = \frac{A_m}{A_c} \times 100$$

$$\% \text{modulation} = \frac{20}{80} \times 100$$

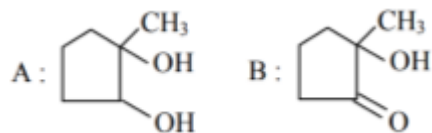
$$\% \text{modulation} = 25\%$$

Ans. 25

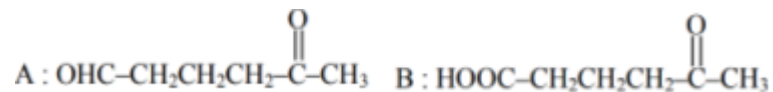
Q31: Identify products A and B.



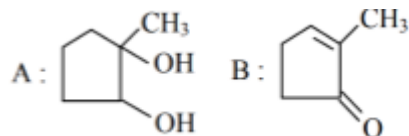
(A)



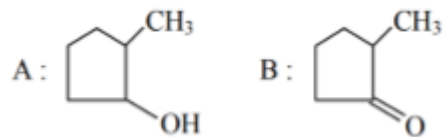
(B)



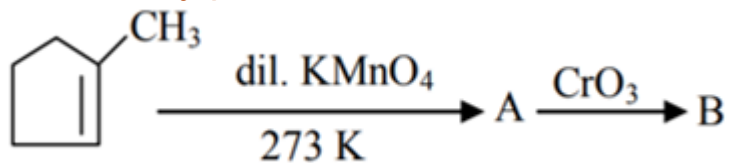
(C)



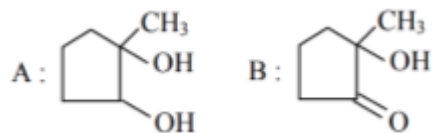
(D)



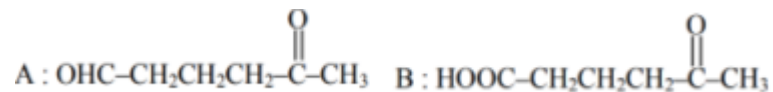
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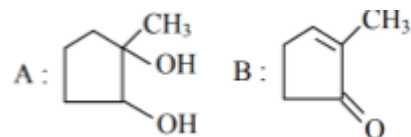
(A)



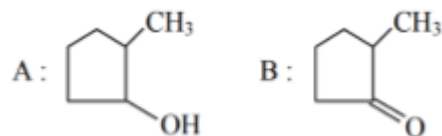
(B)



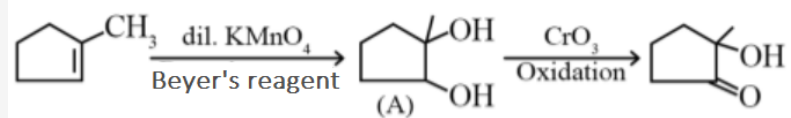
(C)



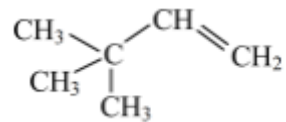
(D)



Solution:

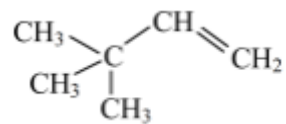


Q32: What is the major product formed by HI on reaction with

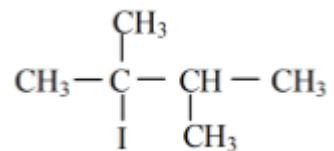


- (A)
$$\begin{array}{c} \text{CH}_3 \\ | \\ \text{CH}_3 - \text{C} - \text{CH} - \text{CH}_3 \\ | \quad | \\ \text{I} \quad \text{CH}_3 \end{array}$$
- (B)
$$\begin{array}{c} \text{CH}_3 - \text{CH} - \text{CH} - \text{CH}_2 - \text{CH}_3 \\ | \quad | \\ \text{CH}_3 \quad \text{I} \end{array}$$
- (C)
$$\begin{array}{c} \text{CH}_3 \\ | \\ \text{CH}_3 - \text{C} - \text{CH} - \text{CH}_3 \\ | \quad | \\ \text{CH}_3 \quad \text{I} \end{array}$$
- (D)
$$\begin{array}{c} \text{CH}_3 \\ | \\ \text{CH}_3 - \text{C} - \text{CH} - \text{CH}_2\text{I} \\ | \quad | \\ \text{CH}_3 \quad \text{H} \end{array}$$

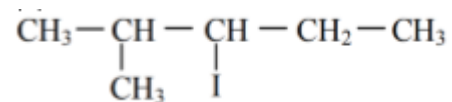
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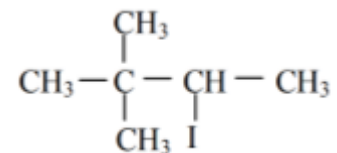
(A)



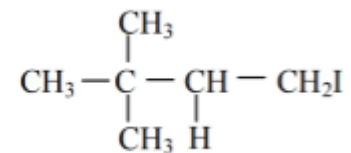
(B)



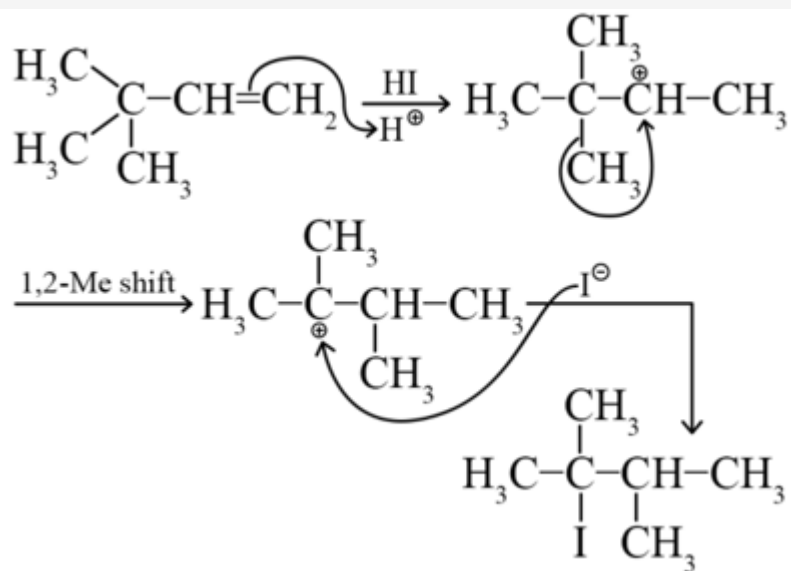
(C)



(D)



Solution:



Q33: Consider the elements Mg, Al, S, P and Si, the correct increasing order of their first ionization enthalpy is

- (A) $\text{Al} < \text{Mg} < \text{Si} < \text{S} < \text{P}$
- (B) $\text{Mg} < \text{Al} < \text{Si} < \text{S} < \text{P}$
- (C) $\text{Mg} < \text{Al} < \text{Si} < \text{P} < \text{S}$
- (D) $\text{Al} < \text{Mg} < \text{S} < \text{Si} < \text{P}$

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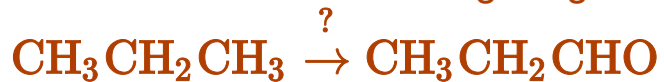
- (A) $\text{Al} < \text{Mg} < \text{Si} < \text{S} < \text{P}$
- (B) $\text{Mg} < \text{Al} < \text{Si} < \text{S} < \text{P}$
- (C) $\text{Mg} < \text{Al} < \text{Si} < \text{P} < \text{S}$
- (D) $\text{Al} < \text{Mg} < \text{S} < \text{Si} < \text{P}$

Solution:

Ionization energy increases along the period generally & ionization energy is directly proportional to fully filled / half-filled configuration.

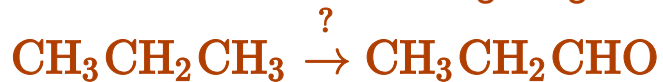
By using the above rule option (i) is correct.

Q34: Which of the following reagent is used for the following reaction?



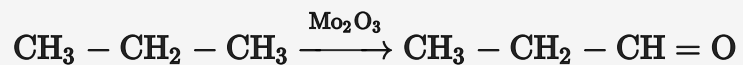
- (A) Copper at high temperature and pressure
- (B) Molybdenum oxide
- (C) Manganese acetate
- (D) Potassium permanganate

Q34: Which of the following reagent is used for the following reaction?



- (A) Copper at high temperature and pressure
- (B) Molybdenum oxide
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- (D) Potassium permanganate

Solution:



The reagent used will be Mo_2O_3

Q35: Which of the following ore is concentrated using group 1 cyanide salt?

- (A) Calamine
- (B) Sphalerite
- (C) Siderite
- (D) Malachite

Q35: Which of the following ore is concentrated using group 1 cyanide salt?

- (A) Calamine
- (B) Sphalerite
- (C) Siderite
- (D) Malachite

Solution:

Sphalerite ore : ZnS

Calamine ore : ZnCO_3

Siderite ore : FeCO_3

Malachite ore : $\text{Cu}(\text{OH})_2 \cdot \text{CuCO}_3$

It is possible to separate two sulphide ores by adjusting proportion of oil to water or by using 'depressants'. In case of an ore containing ZnS and PbS , the depressant used is NaCN .

Q36: Which of the following are isostructural pairs?



(A) I and III only

(B) I and II only

(C) II and III only

(D) III and IV only

Q36: Which of the following are isostructural pairs?



(A) I and III only

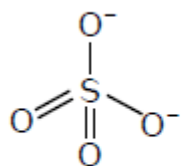
(B) I and II only

(C) II and III only

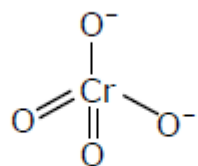
(D) III and IV only

Solution:

(1) SO_4^{2-} and CrO_4^{2-} both have tetrahedral structure.

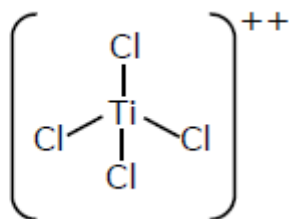
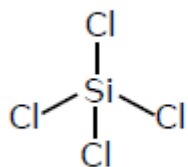


Tetrahedral

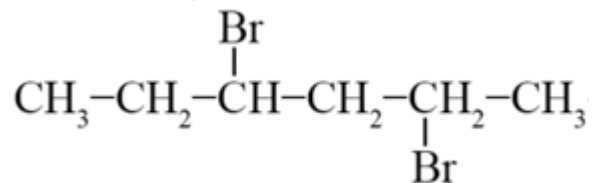


Tetrahedral

(2) SiCl_4 and TiCl_4 both have Tetrahedral structure also.



Q37: The product formed in the first step of the reaction of

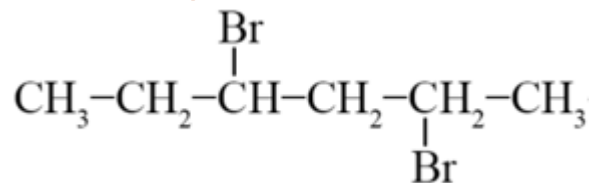


with excess

Mg/Et₂O (Et = C₂H₅) is:

- (A)
$$\text{CH}_3\text{CH}_2-\overset{\text{MgBr}}{\underset{|}{\text{CH}}}-\text{CH}_2-\underset{\text{MgBr}}{\underset{|}{\text{CH}}}-\text{CH}_3$$
- (B)
$$\begin{array}{c} \text{CH}_3-\text{CH}_2-\text{CH}-\text{CH}_2-\text{CH}-\text{CH}_3 \\ | \qquad \qquad | \\ \text{CH}_3-\text{CH}-\text{CH}_2-\text{CH}-\text{CH}_2-\text{CH}_3 \end{array}$$
- (C)
$$\begin{array}{c} \text{CH}_3-\text{CH} \begin{array}{l} \nearrow \text{CH}_2 \\ \searrow \text{CH}-\text{CH}_3 \end{array} \end{array}$$
- (D)
$$\begin{array}{c} \text{CH}_3-\text{CH}_2-\text{CH}-\text{CH}_2-\text{CH}-\text{CH}_3 \\ | \qquad \qquad | \\ \text{CH}_3-\text{CH}_2-\text{CH}-\text{CH}_2-\text{CH}-\text{CH}_3 \end{array}$$

Q37: The product formed in the first step of the reaction of

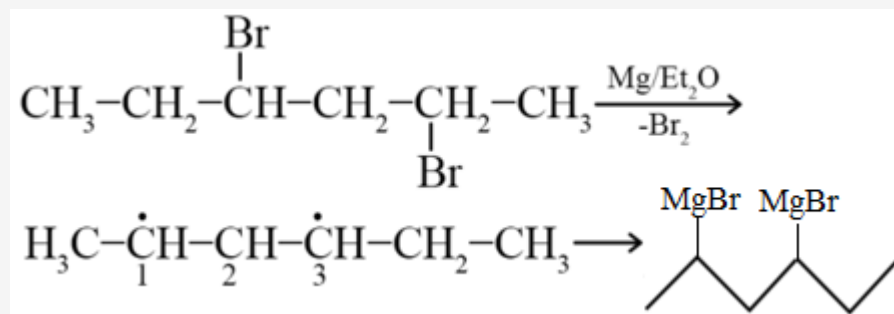


with excess

Mg/Et₂O (Et = C₂H₅) is:

- (A)
$$\text{CH}_3\text{CH}_2-\overset{\text{MgBr}}{\underset{|}{\text{CH}}}-\text{CH}_2-\underset{\text{MgBr}}{\underset{|}{\text{CH}}}-\text{CH}_3$$
- (B)
$$\begin{array}{c} \text{CH}_3-\text{CH}_2-\text{CH}-\text{CH}_2-\text{CH}-\text{CH}_3 \\ | \qquad \qquad | \\ \text{CH}_3-\text{CH}-\text{CH}_2-\text{CH}-\text{CH}_2-\text{CH}_3 \end{array}$$
- (C)
$$\begin{array}{c} \text{CH}_3-\text{CH} \begin{array}{l} \nearrow \text{CH}_2 \\ \searrow \text{CH}-\text{CH}_3 \end{array} \end{array}$$
- (D)
$$\begin{array}{c} \text{CH}_3-\text{CH}_2-\text{CH}-\text{CH}_2-\text{CH}-\text{CH}_3 \\ | \qquad \qquad | \\ \text{CH}_3-\text{CH}_2-\text{CH}-\text{CH}_2-\text{CH}-\text{CH}_3 \end{array}$$

Solution:



Q38: Match List-I with List-II.

	List I		List II
A	Caprolactum	(i)	Natural rubber
B	2-Chloro-1, 3-butadiene	(ii)	Buna-N
C	Isoprene	(iii)	Nylon 6
D	Acrylonitrile	(iv)	Neoprene

Choose the correct answer from the options given below

- (A) A – (iii), B – (iv), C – (i), D – (ii)
- (B) A – (ii), B – (i), C – (iv), D – (iii)
- (C) A – (iv), B – (iii), C – (ii), D – (i)
- (D) A – (i), B – (ii), C – (iii), D – (iv)

Q38: Match List-I with List-II.

	List I		List II
A	Caprolactum	(i)	Natural rubber
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Choose the correct answer from the options given below

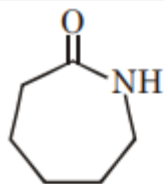
(A) A – (iii), B – (iv), C – (i), D – (ii)

(B) A – (ii), B – (i), C – (iv), D – (iii)

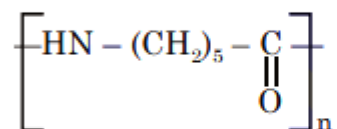
(C) A – (iv), B – (iii), C – (ii), D – (i)

(D) A – (i), B – (ii), C – (iii), D – (iv)

Solution:



Caprolactum is the monomeric unit of polymer Nylon-6



(b) 2-Chlorobuta-1, 3-diene is the monomeric unit of polymer neoprene.

(c) 2-Methylbuta-1, 3-diene is the monomeric unit of polymer natural rubber.

(d) $\text{CH}_2 = \text{CH} - \text{CN}$ (Acrylonitrile) is the one of the monomeric unit of polymer Buna-N

Q39: The gas released during anaerobic degradation of vegetation may lead to

- (A) Global warming and cancer
- (B) Acid rain
- (C) Corrosion of metals
- (D) Ozone hole

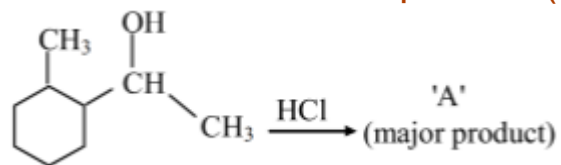
Q39: The gas released during anaerobic degradation of vegetation may lead to

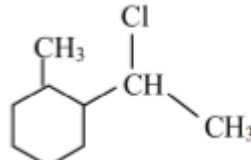
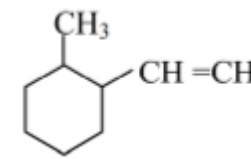
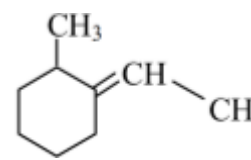
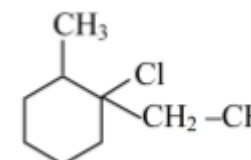
- (A) Global warming and cancer
- (B) Acid rain
- (C) Corrosion of metals
- (D) Ozone hole

Solution:

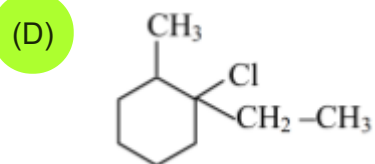
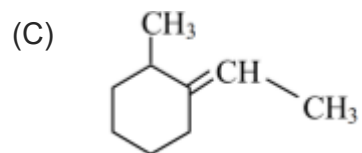
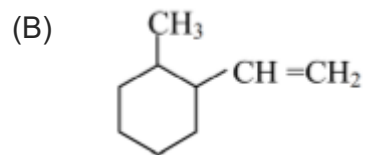
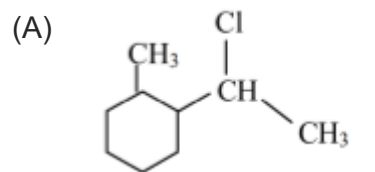
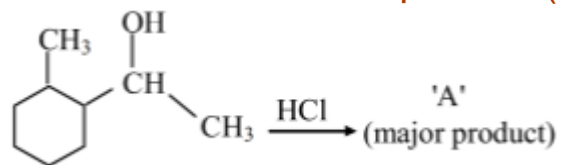
The gas CH_4 evolved due to the anaerobic degradation of vegetation which causes global warming and cancer.

Q40: What is the final product (major) 'A' in the given reaction?

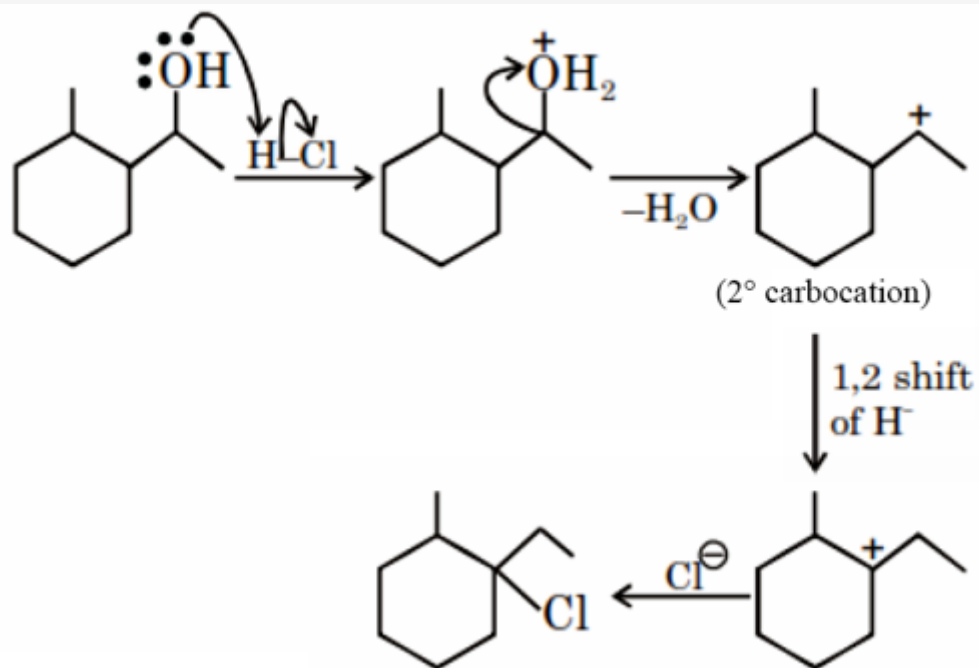


- (A) 
- (B) 
- (C) 
- (D) 

Q40: What is the final product (major) 'A' in the given reaction?



Solution:



Q41: Out of the following, which type of interaction is responsible for the stabilisation of α -helix structure of proteins?

- (A) Ionic bonding
- (B) Vander Waals forces
- (C) Hydrogen bonding
- (D) Covalent bonding

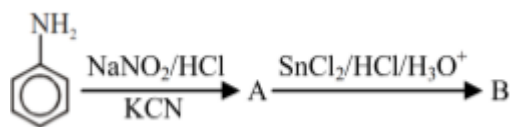
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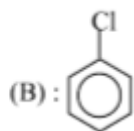
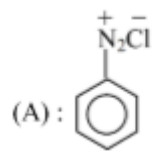
Solution:

Hydrogen bonding is responsible for the stabilization of alpha-helix structure of proteins.

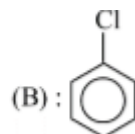
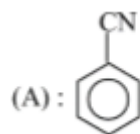
Q42: 'A' and 'B' in the following reactions are



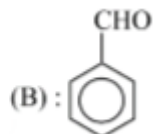
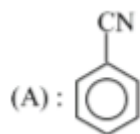
(A)



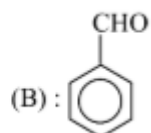
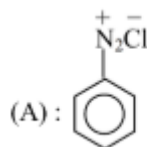
(B)



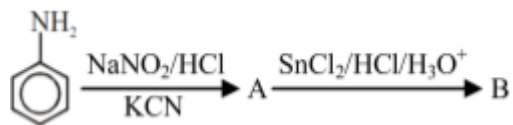
(C)



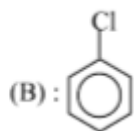
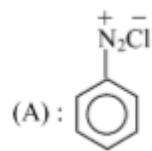
(D)



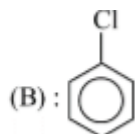
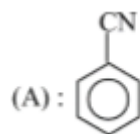
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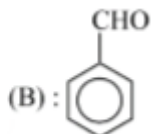
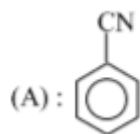
(A)



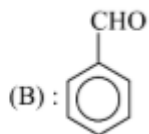
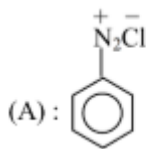
(B)



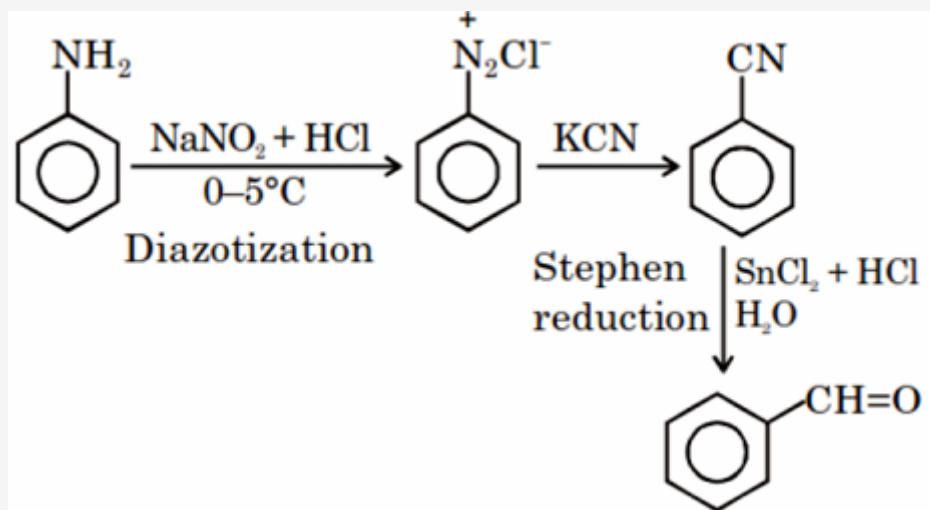
(C)



(D)



Solution:



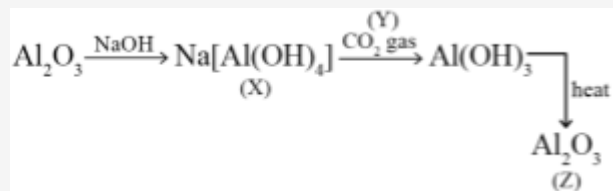
Q43: Al_2O_3 was leached with alkali to get X. The solution of X on passing of gas Y, forms Z, X, and Z respectively are

- (A) $\text{X} = \text{Na}[\text{Al}(\text{OH})_4]$, $\text{Y} = \text{SO}_2$, $\text{Z} = \text{Al}_2\text{O}_3$
- (B) $\text{X} = \text{Al}(\text{OH})_3$, $\text{Y} = \text{SO}_2$, $\text{Z} = \text{Al}_2\text{O}_3 \cdot x\text{H}_2\text{O}$
- (C) $\text{X} = \text{Al}(\text{OH})_3$, $\text{Y} = \text{CO}_2$, $\text{Z} = \text{Al}_2\text{O}_3$
- (D) $\text{X} = \text{Na}[\text{Al}(\text{OH})_4]$, $\text{Y} = \text{CO}_2$, $\text{Z} = \text{Al}_2\text{O}_3 \cdot x\text{H}_2\text{O}$

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- (D) $\text{X} = \text{Na}[\text{Al}(\text{OH})_4]$, $\text{Y} = \text{CO}_2$, $\text{Z} = \text{Al}_2\text{O}_3 \cdot x\text{H}_2\text{O}$

Solution:





Choose the correct option.

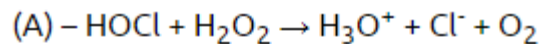
- (A) H_2O_2 acts as reducing agent in equations (A) and (B)
- (B) H_2O_2 acts as reducing and oxidising agent respectively in equations (A) and (B)
- (C) H_2O_2 acts as oxidising agent in equations (A) and (B)
- (D) H_2O_2 act as oxidizing and reducing agent respectively in equations (A) and (B)



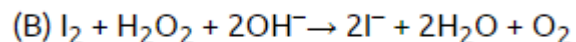
Choose the correct option.

- (A) H_2O_2 acts as reducing agent in equations (A) and (B)
- (B) H_2O_2 acts as reducing and oxidising agent respectively in equations (A) and (B)
- (C) H_2O_2 acts as oxidising agent in equations (A) and (B)
- (D) H_2O_2 act as oxidizing and reducing agent respectively in equations (A) and (B)

Solution:



In this equation, H_2O_2 is reducing chlorine from +1 to -1.

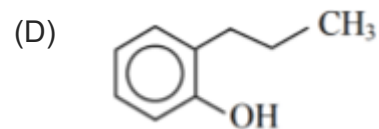
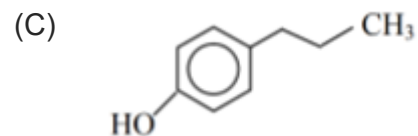
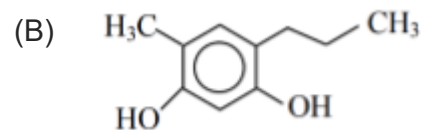
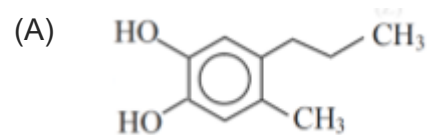


In this equation, H_2O_2 is reducing iodine from 0 to -1.

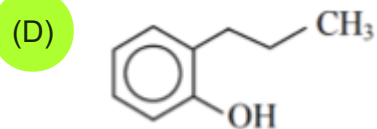
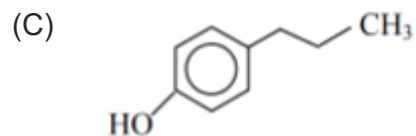
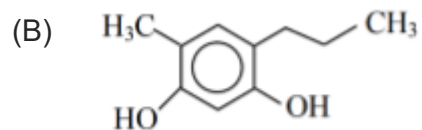
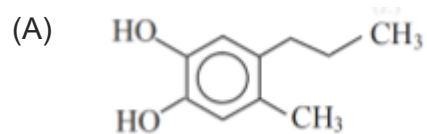
In (A) reduction of HOCl occurs so it will be an oxidising agent hence H_2O_2 will be a reducing agent.

In (B) reduction of I_2 occurs so it will be an oxidising agent and H_2O_2 will be a reducing agent.

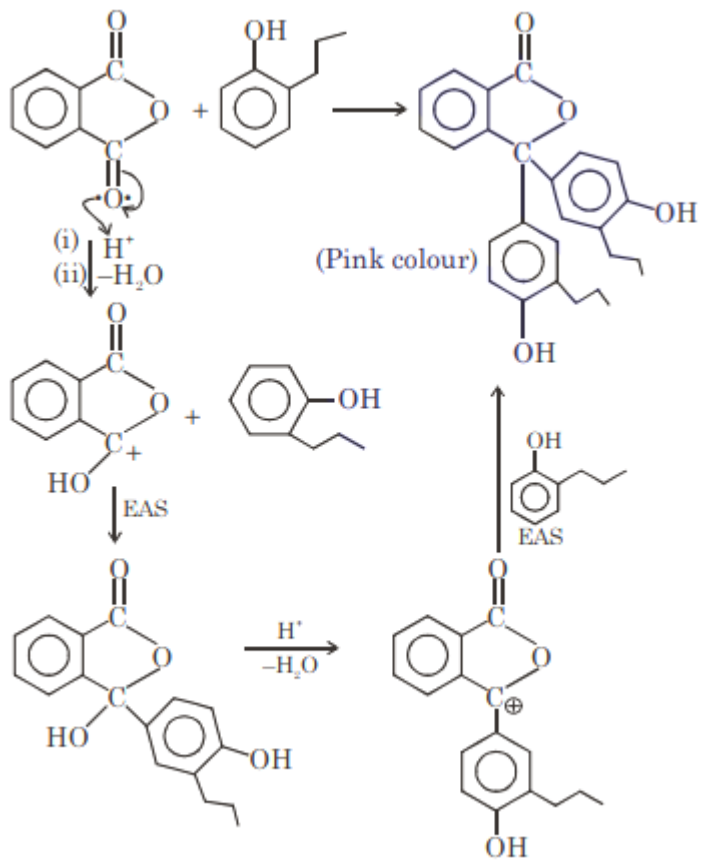
Q45: Which of the following compound gives pink colour on reaction with phthalic anhydride in conc. H_2SO_4 followed by treatment with NaOH ?



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Solution:

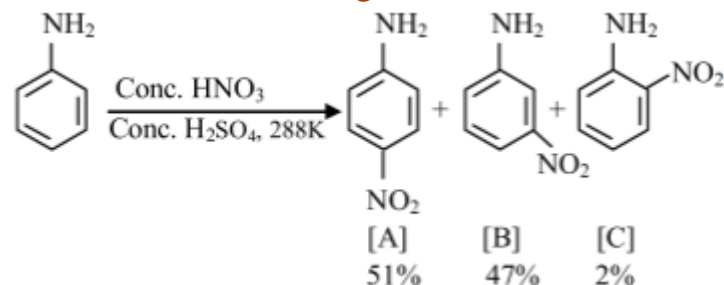


Q46: In the following reaction the reason why *meta*-nitro product also formed is



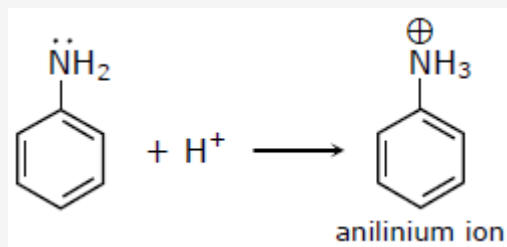
- (A) Low temperature
- (B) $-\text{NH}_2$ group is highly *meta*-directive
- (C) Formation of anilinium ion
- (D) $-\text{NO}_2$ substitution always takes place at *meta*-position

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- (A) Low temperature
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- (C) Formation of anilinium ion
- (D) $-\text{NO}_2$ substitution always takes place at *meta*-position

Solution:



In acidic medium the $-\text{NH}_2$ group in aniline converts into anilinium ion which is meta directing

Q47: The major components in “Gun Metal” are

- (A) Cu, Zn and Ni
- (B) Al, Cu, Mg and Mn
- (C) Cu, Ni and Fe
- (D) Cu, Sn and Zn

Q47: The major components in “Gun Metal” are

- (A) Cu, Zn and Ni
- (B) Al, Cu, Mg and Mn
- (C) Cu, Ni and Fe
- (D) Cu, Sn and Zn

Solution:

"Gun metal" is an alloy of copper with tin and zinc.

Q48: Given below are two statements:

Statement I: Colourless cupric metaborate is reduced to cuprous metaborate in a luminous flame.

Statement II: Cuprous metaborate is obtained by heating boric anhydride and copper sulphate in a non-luminous flame.

In the light of the above statements, choose the most appropriate answer from the options given below.

- (A) Both Statement I and Statement II are false
- (B) Statement I is false but Statement II is true
- (C) Statement I is true but Statement II is false
- (D) Both Statement I and Statement II are true

Q48: Given below are two statements:

Statement I: Colourless cupric metaborate is reduced to cuprous metaborate in a luminous flame.

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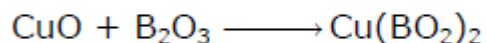
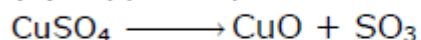
In the light of the above statements, choose the most appropriate answer from the options given below.

- (A) Both Statement I and Statement II are false
- (B) Statement I is false but Statement II is true
- (C) Statement I is true but Statement II is false
- (D) Both Statement I and Statement II are true

Solution:

Both are False

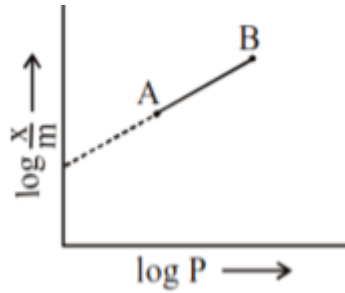
(1) Copper sulphate form copper meta boric with boric anhydride



blue in cold oxidising flame (non luminous flame)

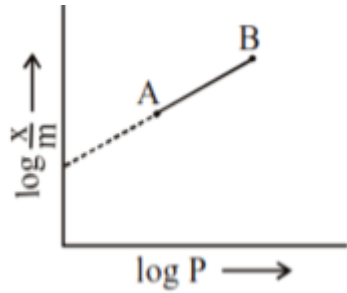
(2) Blue coloured metal borate is reduced to copper in a luminous flame.

Q49: In Freundlich adsorption isotherm, slope of AB line is:



- (A) $\log n$ with $(n > 1)$
- (B) n with $(n, 0.1 \text{ to } 0.5)$
- (C) $\log \frac{1}{n}$ with $(n < 1)$
- (D) $\frac{1}{n}$ with $(\frac{1}{n} = 0 \text{ to } 1)$

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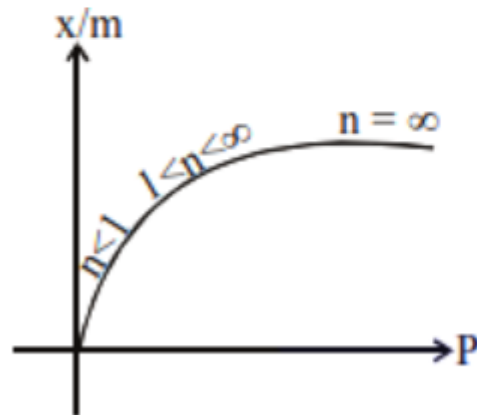
Solution:

$$\frac{x}{m} = K(P)^{1/n}$$

$$\log\left(\frac{x}{m}\right) = \log K + \frac{1}{n} \log P$$

$$y = c + mx$$

$m = 1/n$ so slope will be equal to $1/n$.



Hence $\boxed{0 \leq \frac{1}{n} \leq 1}$

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- (A) Fe
- (B) Co
- (C) Zn
- (D) Cu

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- (B) Co
- (C) Zn
- (D) Cu

Solution:

Only copper shows a positive value for electrode potential of M^{2+}/M of 3d-series elements.

Q51: Gaseous cyclobutene isomerizes to butadiene in a first order process which has a 'k' value of $33 \times 10^{-4} \text{s}^{-1}$ at 153°C . The time in minutes it takes for the isomerization to proceed 40% to completion at this temperature is _____. (Rounded off to the nearest integer)

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Solution:

For first order Rxn :-

$$t = \frac{2.303}{k} \log \left[\frac{100}{100 - x} \right]$$

$$X = 40, k = 3.3 \times 10^{-4}$$

$$t = \frac{2.303}{3.3 \times 10^{-4}} \log \left[\frac{100}{60} \right]$$

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$$t = \frac{2.303}{3.3 \times 10^{-4}} \times 0.22$$

$$t = 0.1535.3 \times 10^4$$

$$t = 1535 \text{ sec.}$$

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$$t = 1535 \text{ sec} = 25.6 \text{ Min.}$$

Q52: 4.5 g of compound A (MW = 90) was used to make 250 mL of its aqueous solution. The molarity of the solution in M is $x \times 10^{-1}$. The value of x is _____. (Rounded off to the nearest integer)

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Solution:

$$\text{Moles of A} = \frac{\text{Weight}}{\text{M.w}}$$

$$= \frac{4.5}{90} = \frac{1}{20} = 0.05$$

$$\text{Volume (Lit)} = \frac{250}{1000} = 0.250 \text{ lit}$$

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$$\text{Molarity (M)} = \frac{\text{Mole}}{(\text{Lit})\text{volume}} = \frac{0.05}{0.250} = 0.2$$

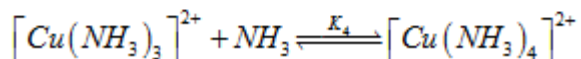
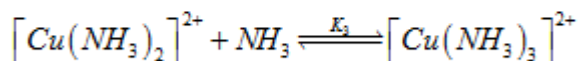
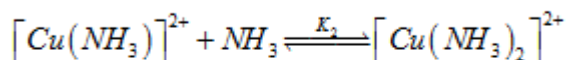
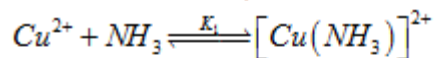
$$= 2 \times 10^{-1} \frac{\text{mol}}{\text{Lit}}$$

$$x = 2$$

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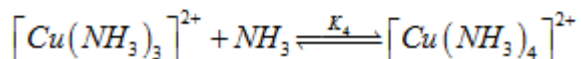
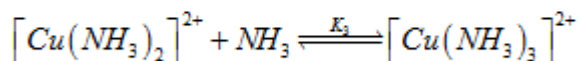
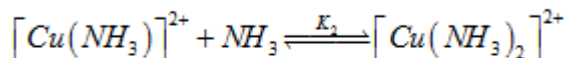
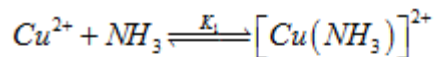
Q53: The stepwise formation of $[\text{Cu}(\text{NH}_3)_4]^{2+}$ is given below:



The value of stability constants K_1 , K_2 , K_3 and K_4 are 10^4 , 1.58×10^3 , 5×10^2 and 10^2 respectively. The overall equilibrium constants for dissociation of $[\text{Cu}(\text{NH}_3)_4]^{2+}$ is $x \times 10^{-12}$.

The value of x is _____. (rounded off to the nearest integer)

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The value of x is _____. (rounded off to the nearest integer)

Solution:

$$[\text{Cu}(\text{NH}_3)_4]^{2+} \rightleftharpoons \text{Cu}^{2+} + 4\text{NH}_3$$

$$K = \frac{1}{K_1 K_2 K_3 K_4} = \frac{1}{10^4 \times 1.58 \times 10^3 \times 5 \times 10^2 \times 10^2}$$

$$= 1.26 \times 10^{-12} = 1.26 \approx 1$$

Q54: Number of amphoteric compounds among the following is _____.

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2.00

Solution:

$\text{Be}(\text{OH})_2$, BeO are amphoteric compounds

Q55: When 9.45 g of ClCH_2COOH is added to 500 mL of water, its freezing point drops by 0.5°C . The dissociation constant of ClCH_2COOH is $x \times 10^{-3}$. The value of x is _____.

(Rounded off to the nearest integer)

$$[K_{f(\text{H}_2\text{O})} = 1.86 \text{ K kg mol}^{-1}]$$

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$$[K_{f(\text{H}_2\text{O})} = 1.86 \text{ K kg mol}^{-1}]$$

Solution:



t = 0	C	O	O
t = t	C - cα	cα	cα

Total no. of moles = c + cα = c(1 + α)

$$i = \frac{\text{observed}}{\text{calculate}} = \frac{c(1 + \alpha)}{c} = (1 + \alpha)$$

M.W. = 94.5

$$\Delta T_f = i \times k_f \times m$$

$$\Delta T_f = 0.5^\circ\text{C}$$

$$i = 1 + \alpha$$

$$0.5 = (1 + \alpha) \times 1.86 \times \frac{9.45}{\frac{94.5}{\frac{500}{1000}}}$$

$$m = \frac{\text{mole}}{\text{k.g(Solvent)}}$$

$$k_t = 1.86 \text{ k kg/mol}$$

$$(1 + \alpha) = \frac{2.5}{1.86}$$

$$\alpha = \frac{0.64}{1.86} = \frac{32}{93}$$

$$K_a = \frac{C\alpha^2}{1 - \alpha} = \frac{0.2 \times 1024}{93 \times 93 \times \frac{61}{93}}$$

$$K_a = 0.0351 = 35.1 \times 10^{-3}$$

Q56: The coordination number of an atom in a body-centered cubic structure is _____.

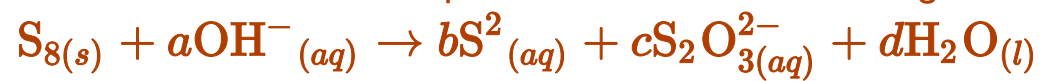
[Assume that the lattice is made up of atoms.]

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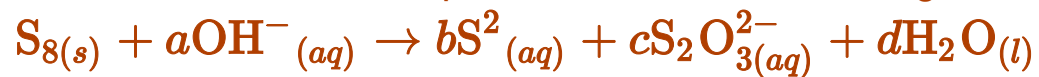
Coordination number of an atom in BCC structure is 8.

Q57: The reaction of sulphur in alkaline medium is given below:



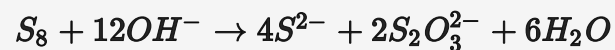
The values of 'a' is _____. (Integer answer)

Q57: The reaction of sulphur in alkaline medium is given below:



The values of 'a' is _____. (Integer answer)

Solution:



Q58: For the reaction $A_{(g)} \rightarrow B_{(g)}$, the value of the equilibrium constant at 300 K and 1 atm is equal to 100.0. The value of $\Delta_r G^0$ for the reaction at 300 K and 1 atm in J mol^{-1} is $-xR$, where x is _____. (Rounded off to the nearest integer)

[$R = 8.31 \text{ J mol}^{-1}\text{K}^{-1}$ and $\ln 10 = 2.3$)

Q58: For the reaction $A_{(g)} \rightarrow B_{(g)}$, the value of the equilibrium constant at 300 K and 1 atm is equal to 100.0. The value of $\Delta_r G^\circ$ for the reaction at 300 K and 1 atm in J mol^{-1} is $-xR$, where x is _____. (Rounded off to the nearest integer)

[$R = 8.31 \text{ J mol}^{-1}\text{K}^{-1}$ and $\ln 10 = 2.3$)]

Solution:

$$\begin{aligned}\Delta_r G^\circ &= -RT \ln K_{\text{eq}} \\ &= -R \times 300 \times 2 \times 2.3 \\ &= -1380 R\end{aligned}$$

Q59: A proton and a Li^{3+} nucleus is accelerated by the same potential. If λ_{Li} and λ_p denote the de Broglie wavelengths of Li^{3+} and proton respectively, then the value $\frac{\lambda_{Li}}{\lambda_p}$ is $x \times 10^{-1}$. The value of x is _____. (Rounded off to the nearest integer)

[Mass of $\text{Li}^{3+} = 8.3$ mass of proton]

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[Mass of $\text{Li}^{3+} = 8.3$ mass of proton]

2.00

Solution:

$$\lambda_{DB} \propto \frac{1}{\sqrt{2mK.E.}}$$

$$\frac{\lambda_{Li^{3+}}}{\lambda_p} = \sqrt{\frac{m_p \times e_p V}{8.3m_p \times 3e_p V}}$$

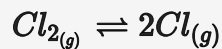
$$= \sqrt{\frac{1}{25}} = \frac{1}{5} = 0.2 = 2 \times 10^{-1}$$

Q60: At 1990 K and 1 atm pressure, there are equal number of Cl_2 molecules and Cl atoms in the reaction mixture. The value of K_p for the reaction $\text{Cl}_{2(g)} \rightleftharpoons 2\text{Cl}_{(g)}$ under the above conditions is $x \times 10^{-1}$. The value of x is _____. (Rounded off to the nearest integer)

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5.00

Solution:



Moles x x

at eq^m

P.P $\frac{1}{2}$ $\frac{1}{2}$

$$K_P = \frac{P_{\text{Cl}}^2}{P_{\text{Cl}_2}}$$

$$= \frac{\left(\frac{1}{2}\right)^2}{\frac{1}{2}} = \frac{1}{2} = 0.5$$

$$= 5 \times 10^{-1}$$

$$x = 5$$

Q61: The distance of the point $(1, 1, 9)$ from the point of intersection of the line

$\frac{x-3}{1} = \frac{y-4}{2} = \frac{z-5}{2}$ and the plane $x + y + z = 17$ is

(A) $2\sqrt{19}$

(B) 38

(C) $19\sqrt{2}$

(D) $\sqrt{38}$

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(A) $2\sqrt{19}$

(B) 38

(C) $19\sqrt{2}$

(D) $\sqrt{38}$

Solution:

$$\frac{x-3}{1} = \frac{y-4}{2} = \frac{z-5}{2} = \lambda$$

$$\Rightarrow x = \lambda + 3, y = 2\lambda + 4, z = 2\lambda + 5$$

Which lies on given plane hence

$$\Rightarrow \lambda + 3 + 2\lambda + 4 + 2\lambda + 5 = 17$$

$$\Rightarrow \lambda = \frac{5}{5} = 1$$

Hence, point of intersection is Q (4, 6, 7)

∴ Required distance = PQ

$$= \sqrt{9 + 25 + 4}$$

$$= \sqrt{38}$$

Q62: The locus of the mid-point of the line segment joining the focus of the parabola $y^2 = 4ax$ to a moving point of the parabola, is another parabola whose directrix is

(A) $x = \frac{a}{2}$

(B) $x = 0$

(C) $x = -\frac{a}{2}$

(D) $x = a$

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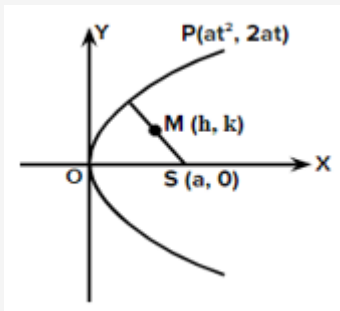
(B) $x = 0$

(C) $x = -\frac{a}{2}$

(D) $x = a$

Solution:

$$h = \frac{at^2+a}{2}, k = \frac{2at+0}{2}$$
$$\Rightarrow t^2 = \frac{2h-a}{a} \text{ and } t = \frac{k}{a}$$



$$\Rightarrow \frac{k^2}{a^2} = \frac{2h-a}{a}$$

$$\Rightarrow \text{Locus of } (h, k) \text{ is } y^2 = a(2x - a)$$

$$\Rightarrow y^2 = 2a\left(x - \frac{a}{2}\right)$$

$$\text{Its directrix is } x - \frac{a}{2} = -\frac{a}{2} \Rightarrow x = 0$$

Q63: If the tangent to the curve $y = x^3$ at the point $P(t, t^3)$ meets the curve again at Q, then the ordinate of the point which divides PQ internally in the ratio 1 : 2 is

(A) $-2t^3$

(B) $2t^3$

(C) $-t^3$

(D) 0

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(B) $2t^3$

(C) $-t^3$

(D) 0

Solution:

Equation of tangent at $P(t, t^3)$

$$(y - t^3) = 3t^2(x - t) \quad \dots (i)$$

now solve the above equation with

$$y = x^3 \quad \dots (ii)$$

By (i) & (ii)

$$x^3 - t^3 = 3t^2(x - t)$$

$$x^2 + xt + t^2 = 3t^2$$

$$x^2 + xt - 2t^2 = 0$$

$$(x - t)(x + 2t) = 0$$

$$\Rightarrow x = -2t$$

$$\Rightarrow Q(-2t, -8t^3)$$

$$\text{Ordinate of required point} = \frac{2t^3 + (-8t^3)}{3} = -2t^3$$

Q64: The function $f(x) = \frac{4x^3 - 3x^2}{6} - 2 \sin x + (2x - 1) \cos x$:

- (A) Decreases in $(-\infty, \frac{1}{2}]$
- (B) Increases in $(-\infty, \frac{1}{2}]$
- (C) Increases in $[\frac{1}{2}, \infty)$
- (D) Decreases in $[\frac{1}{2}, \infty)$

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(D) Decreases in $[\frac{1}{2}, \infty)$

Solution:

$$f'(x) = (2x - 1)(x - \sin x)$$

$$\Rightarrow f'(x) \geq 0 \text{ for } x \in (-\infty, 0] \cup [\frac{1}{2}, \infty)$$

$$\text{and } f'(x) \leq 0 \text{ for } x \in [0, \frac{1}{2}]$$

Q65: The population $P = P(t)$ at time ' t ' of a certain species follows the differential equation $\frac{dP}{dt} = 0.5P - 450$. If $P(0) = 850$, then the time at which population becomes zero is

- (A) $\log_e 9$
- (B) $\frac{1}{2} \log_e 18$
- (C) $\log_e 18$
- (D) $2 \log_e 18$

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- (B) $\frac{1}{2}\log_e 18$
- (C) $\log_e 18$
- (D) $2\log_e 18$

Solution:

$$\frac{dP(t)}{dt} = \frac{P(t) - 900}{2}$$

$$\int_0^t \frac{dP(t)}{P(t) - 900} = \int_0^t \frac{dt}{2}$$

$$\{\log_e |P(t) - 900|\}_0^t = \left\{ \frac{t}{2} \right\}_0^t$$

$$\log_e |P(t) - 900| - \log_e |P(0) - 900| = \frac{t}{2}$$

$$\log_e |P(t) - 900| - \log_e 50 = \frac{t}{2}$$

Let $t = t_1$, $P(t) = 0$ hence

$$\log_e |P(t) - 900| - \log_e 50 = \frac{t_1}{2}$$

$$t_1 = 2\log_e 18$$

Q66: If $\int \frac{\cos x - \sin x}{\sqrt{8 - \sin 2x}} dx = a \sin^{-1} \left(\frac{\sin x + \cos x}{b} \right) + c$, where c is a constant of integration, then the ordered pair (a, b) is equal to

- (A) $(3, 1)$
- (B) $(1, -3)$
- (C) $(1, 3)$
- (D) $(-1, 3)$

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- (A) $(3, 1)$
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Solution:

$$\text{Put } \sin x + \cos x = t$$

$$\Rightarrow 1 + \sin 2x = t^2$$

$$\Rightarrow (\cos x - \sin x) dx = dt$$

$$\therefore I = \int \frac{dt}{\sqrt{8 - (t^2 - 1)}} = \int \frac{dt}{\sqrt{9 - t^2}}$$

$$I = \sin^{-1} \left(\frac{t}{3} \right) + C$$

$$I = \sin^{-1} \left(\frac{\sin x + \cos x}{3} \right) + C$$

$$\Rightarrow a = 1 \text{ and } b = 3$$

Q67: The area (in sq. units) of the part of the circle $x^2 + y^2 = 36$, which is outside the parabola $y^2 = 9x$, is

- (A) $24\pi + 3\sqrt{3}$
- (B) $12\pi - 3\sqrt{3}$
- (C) $12\pi + 3\sqrt{3}$
- (D) $24\pi - 3\sqrt{3}$

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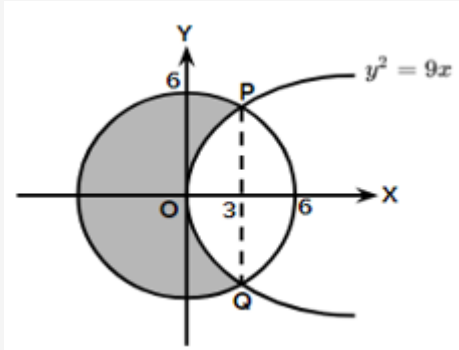
(B) $12\pi - 3\sqrt{3}$

(C) $12\pi + 3\sqrt{3}$

(D) $24\pi - 3\sqrt{3}$

Solution:

The curves intersect at points $(3, \pm 3\sqrt{3})$



Required area

$$\begin{aligned}
 &= \pi r^2 - 2 \left[\int_0^3 \sqrt{9x} dx + \int_3^6 \sqrt{36 - x^2} dx \right] \\
 &= 36\pi - 12\sqrt{3} - 2 \left| \frac{x}{2} \sqrt{36 - x^2} + 18 \sin^{-1} \left(\frac{x}{6} \right) \right|_3^6 \\
 &= 36\pi - 12\sqrt{3} - 2 \left(9\pi - \left(\frac{9\sqrt{3}}{2} + 3\pi \right) \right) \\
 &= 24\pi - 3\sqrt{3}
 \end{aligned}$$

Q68: The equation of the plane passing through the point $(1, 2, -3)$ and perpendicular to the planes $3x + y - 2z = 5$ and $2x - 5y - z = 7$, is

(A) $6x - 5y + 2z + 10 = 0$

(B) $6x - 5y + 2z - 2 = 0$

(C) $3x - 10y + 2z + 11 = 0$

(D) $11x + y + 17z + 38 = 0$

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(B) $6x - 5y + 2z - 2 = 0$

(C) $3x - 10y + 2z + 11 = 0$

(D) $11x + y + 17z + 38 = 0$

Solution:

$$\text{Normal vector of required plane is } \vec{n} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & 1 & -2 \\ 2 & -5 & -1 \end{vmatrix} = -11\hat{i} - \hat{j} - 17\hat{k}$$

$$\therefore 11(x - 1) + (y - 2) + 17(z + 3) = 0$$

$$11x + y + 17z + 38 = 0$$

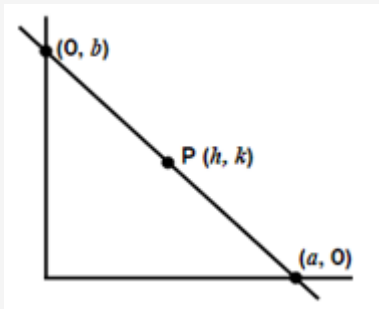
Q69: A man is walking on a straight line. The arithmetic mean of the reciprocals of the intercepts of this line on the coordinate axes is $\frac{1}{4}$. Three stones A, B and C are placed at the points (1, 1), (2, 2) and (4, 4) respectively. Then which of these stones is/are on the path of the man?

- (A) A only
- (B) All the three
- (C) B only
- (D) C only

Q69: A man is walking on a straight line. The arithmetic mean of the reciprocals of the intercepts of this line on the coordinate axes is $\frac{1}{4}$. Three stones A, B and C are placed at the points (1, 1), (2, 2) and (4, 4) respectively. Then which of these stones is/are on the path of the man?

- (A) A only
- (B) All the three
- ☒ (C) B only
- (D) C only

Solution:



$$\frac{x}{a} + \frac{y}{b} = 1$$

$$\frac{h}{a} + \frac{k}{b} = 1 \quad \dots (i)$$

$$\frac{\frac{1}{a} + \frac{1}{b}}{2} = \frac{1}{4}$$

$$\therefore \frac{1}{a} + \frac{1}{b} = \frac{1}{2} \quad \dots (ii)$$

$\therefore$ Lines passes through fixed point (2, 2)
(from (1) and (2))

Q70: If $e^{(\cos^2 x + \cos^4 x + \cos^6 x + \dots \infty) \log_e 2}$ satisfies the equation $t^2 - 9t + 8 = 0$, then the value of $\frac{2 \sin x}{\sin x + \sqrt{3} \cos x} \left(0 < x < \frac{\pi}{2}\right)$ is

(A) $\frac{1}{2}$

(B) $\sqrt{3}$

(C) $2\sqrt{3}$

(D) $\frac{3}{2}$

Q70: If $e^{(\cos^2 x + \cos^4 x + \cos^6 x + \dots \infty) \log_e 2}$ satisfies the equation $t^2 - 9t + 8 = 0$, then the value of $\frac{2 \sin x}{\sin x + \sqrt{3} \cos x} \left(0 < x < \frac{\pi}{2}\right)$ is

(A) $\frac{1}{2}$

(B) $\sqrt{3}$

(C) $2\sqrt{3}$

(D) $\frac{3}{2}$

Solution:

$$e^{(\cos^2 x + \cos^4 x + \cos^6 x + \dots \infty) \log_e 2} = 2^{(\cos^2 x + \cos^4 x + \dots \infty)}$$

$$= 2^{\cot^2 x}$$

Given $t^2 - 9t + 8 = 0 \Rightarrow t = 1, 8$

$\Rightarrow 2^{\cot^2 x} = 1$ and $2^{\cot^2 x} = 8$

$\Rightarrow \cot^2 x = 0, 3$

$0 < x < \frac{\pi}{2} \Rightarrow \cot x = \sqrt{3}$

$\Rightarrow \frac{2 \sin x}{\sin x + \sqrt{3} \cos x} = \frac{2}{1 + \sqrt{3} \cot x} = \frac{2}{4} = \frac{1}{2}$

Q71: $\lim_{x \rightarrow 0} \frac{\int_0^{x^2} (\sin \sqrt{t}) dt}{x^3}$ is equal to

(A) $\frac{1}{15}$

(B) 0

(C) $\frac{2}{3}$

(D) $\frac{3}{2}$

Q71: $\lim_{x \rightarrow 0} \frac{\int_0^{x^2} (\sin \sqrt{t}) dt}{x^3}$ is equal to

(A) $\frac{1}{15}$

(B) 0

(C) $\frac{2}{3}$

(D) $\frac{3}{2}$

Solution:

$$\lim_{x \rightarrow 0^+} \frac{\int_0^{x^2} \sin \sqrt{t} dt}{x^3} = \lim_{x \rightarrow 0^+} \frac{(\sin|x|)2x}{3x^2} = \lim_{x \rightarrow 0^+} \left(\frac{\sin x}{x} \right) \times \frac{2}{3} = \frac{2}{3}$$

Q72: The statement among the following that is a tautology is

(A) $B \rightarrow [A \wedge (A \rightarrow B)]$

(B) $A \vee (A \wedge B)$

(C) $[A \wedge (A \rightarrow B)] \rightarrow B$

(D) $A \wedge (A \vee B)$

Q72: The statement among the following that is a tautology is

(A) $B \rightarrow [A \wedge (A \rightarrow B)]$

(B) $A \vee (A \wedge B)$

(C) $[A \wedge (A \rightarrow B)] \rightarrow B$

(D) $A \wedge (A \vee B)$

Solution:

$$\begin{aligned} & A \wedge (\sim A \vee B) \rightarrow B \\ &= [(A \wedge \sim A) \vee (A \wedge B)] \rightarrow B \\ &= (A \wedge B) \rightarrow B \\ &= \sim A \vee \sim B \vee B \\ &= t \end{aligned}$$

Q73: An ordinary dice is rolled for a certain number of times. If the probability of getting an odd number 2 times is equal to the probability of getting an even number 3 times, then the probability of getting an odd number for odd number of times is

(A) $\frac{1}{32}$

(B) $\frac{5}{16}$

(C) $\frac{3}{16}$

(D) $\frac{1}{2}$

Q73: An ordinary dice is rolled for a certain number of times. If the probability of getting an odd number 2 times is equal to the probability of getting an even number 3 times, then the probability of getting an odd number for odd number of times is

(A) $\frac{1}{32}$

(B) $\frac{5}{16}$

(C) $\frac{3}{16}$

(D) $\frac{1}{2}$

Solution:

$$P(\text{odd no. twice}) = P(\text{even no. thrice})$$

$$\Rightarrow {}^nC_2 \left(\frac{1}{2}\right)^n = {}^nC_3 \left(\frac{1}{2}\right)^n \Rightarrow n = 5$$

$$\text{Success is getting an odd number then } P(\text{Odd successes}) = P(1) + P(3) + P(5)$$

$$\begin{aligned} &= {}^5C_1 \left(\frac{1}{2}\right)^5 + {}^5C_3 \left(\frac{1}{2}\right)^5 + {}^5C_5 \left(\frac{1}{2}\right)^5 \\ &= \frac{16}{2^5} = \frac{1}{2} \end{aligned}$$

Q74: Let p and q be two positive numbers such that $p + q = 2$ and $p^4 + q^4 = 272$. Then p and q are roots of the equation

(A) $x^2 - 2x + 136 = 0$

(B) $x^2 - 2x + 8 = 0$

(C) $x^2 - 2x + 16 = 0$

(D) $x^2 - 2x + 2 = 0$

Q74: Let p and q be two positive numbers such that $p + q = 2$ and $p^4 + q^4 = 272$. Then p and q are roots of the equation

(A) $x^2 - 2x + 136 = 0$

(B) $x^2 - 2x + 8 = 0$

(C) $x^2 - 2x + 16 = 0$

(D) $x^2 - 2x + 2 = 0$

Solution:

$$(p^2 + q^2)^2 - 2p^2q^2 = 272$$

$$\left((p + q)^2 - 2pq\right)^2 - 2p^2q^2 = 272$$

$$16 - 16pq + 2p^2q^2 = 272$$

$$(pq)^2 - 8pq - 128 = 0$$

$$pq = \frac{8 \pm 24}{2} = 16, -8$$

$$pq = 16$$

Q75: If $f : \mathbf{R} \rightarrow \mathbf{R}$ is a function defined by $f(x) = [x - 1] \cos\left(\frac{2x-1}{2}\right)\pi$, where $[.]$ denotes the greatest integer function, then f is

- (A) Discontinuous at all integral values of x except at $x = 1$
- (B) Discontinuous only at $x = 1$
- (C) Continuous only at $x = 1$
- (D) Continuous for every real x

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- (B) Discontinuous only at $x = 1$
- (C) Continuous only at $x = 1$
- (D) Continuous for every real x

Solution:

Doubtful points are $x = n, n \in \mathbf{I}$

$$L.H.L = \lim_{x \rightarrow n^-} [x - 1] \cos\left(\frac{2x-1}{2}\right)\pi = (n-2) \cos\left(\frac{2n-1}{2}\right)\pi = 0$$

$$R.H.L = \lim_{x \rightarrow n^+} [x - 1] \cos\left(\frac{2x-1}{2}\right)\pi = (n-1) \cos\left(\frac{2n-1}{2}\right)\pi = 0$$

$$f(n) = 0$$

Hence continuous

Q76: A scientific committee is to be formed from 6 Indians and 8 foreigners, which includes at least 2 Indians and double the number of foreigners as Indians. Then the number of ways the committee can be formed is

- (A) 575
- (B) 1050
- (C) 1625
- (D) 560

Q76: A scientific committee is to be formed from 6 Indians and 8 foreigners, which includes at least 2 Indians and double the number of foreigners as Indians. Then the number of ways the committee can be formed is

- (A) 575
- (B) 1050
- ☒ (C) 1625
- (D) 560

Solution:

$$\begin{aligned} & (2I, 4F) + (3I, 6F) + (4I, 8F) \\ &= {}^6C_2 {}^8C_4 + {}^6C_3 {}^8C_6 + {}^6C_4 {}^8C_8 \\ &= 15 \times 70 + 20 \times 28 + 15 \times 1 \\ &= 1050 + 560 + 15 = 1625 \end{aligned}$$

Q77: The value of

$$- {}^{15}C_1 + 2 {}^{15}C_2 - 3 {}^{15}C_3 + \cdots - 15 {}^{15}C_{15} + {}^{14}C_1 + {}^{14}C_3 + {}^{14}C_5 + \cdots + {}^{14}C_{11}$$

is

- (A) 2^{14}
- (B) $2^{13} - 13$
- (C) 2^{16}
- (D) $2^{13} - 14$

Q77: The value of

$- {}^{15}C_1 + 2 {}^{15}C_2 - 3 {}^{15}C_3 + \dots - 15 {}^{15}C_{15} + {}^{14}C_1 + {}^{14}C_3 + {}^{14}C_5 + \dots + {}^{14}C_{11}$
is

- (A) 2^{14}
- (B) $2^{13} - 13$
- (C) 2^{16}
- (D) $2^{13} - 14$

Solution:

$$\begin{aligned}
 S_1 &= -{}^{15}C_1 + 2.{}^{15}C_2 - \dots - 15.{}^{15}C_{15} \\
 &= \sum_{r=1}^{15} (-1)^r \cdot r \cdot {}^{15}C_r = 15 \sum_{r=1}^{15} (-1)^r {}^{14}C_{r-1} \\
 &= 15 (-{}^{14}C_0 + {}^{14}C_1 - \dots - {}^{14}C_{14}) = 15 (0) = 0 \\
 S_2 &= {}^{14}C_1 + {}^{14}C_3 + \dots + {}^{14}C_{11} \\
 &= ({}^{14}C_1 + {}^{14}C_3 + \dots + {}^{14}C_{11} + {}^{14}C_{13}) - {}^{14}C_{13} \\
 &= 2^{13} - 14 \\
 S_1 + S_2 &= 2^{13} - 14
 \end{aligned}$$

Q78: The system of linear equations

$$3x - 2y - kz = 10$$

$$2x - 4y - 2z = 6$$

$$x + 2y - z = 5m$$

is inconsistent if:

(A) $k = 3, m = \frac{4}{5}$

(B) $k \neq 3, m \in R$

(C) $k \neq 3, m \neq \frac{4}{5}$

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(C) $k \neq 3, m \neq \frac{4}{5}$

(D) $k = 3, m \neq \frac{4}{5}$

Solution:

$$\Delta = \begin{vmatrix} 3 & -2 & -k \\ 2 & -4 & -2 \\ 1 & 2 & -1 \end{vmatrix}$$
$$\Rightarrow 3(4 + 4) + 2(-2 + 2) - k(4 + 4) = 0$$
$$\Rightarrow k = 3$$

$$\Delta_x = \begin{vmatrix} 10 & -2 & -3 \\ 6 & -4 & -2 \\ 5m & 2 & -1 \end{vmatrix} = 2(4 - 5m)$$

$$\Delta_y = \begin{vmatrix} 3 & 10 & -3 \\ 2 & 6 & -2 \\ 1 & 5m & -1 \end{vmatrix} = 0$$

$$\Delta_z = \begin{vmatrix} 3 & -2 & 10 \\ 2 & -4 & 6 \\ 1 & 2 & 5m \end{vmatrix} = 8(4 - 5m)$$

Hence $k = 3$ and $m \neq \frac{4}{5}$

Q79: Two vertical poles are 150 m apart and the height of one is three times that of the other. If from the middle point of the line joining their feet, an observer finds the angles of elevation of their tops to be complementary, then the height of the shorter pole (in meters) is

- (A) 25
- (B) $25\sqrt{3}$
- (C) 30
- (D) $20\sqrt{3}$

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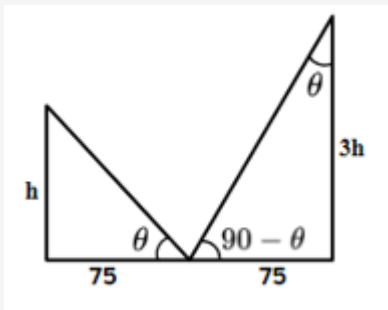
(A) 25

(B) $25\sqrt{3}$

(C) 30

(D) $20\sqrt{3}$

Solution:



$$\tan \theta = \frac{h}{75} = \frac{75}{3h}$$

$$\Rightarrow h^2 = \frac{(75)^2}{3}$$

$$h = 25\sqrt{3} \text{ m}$$

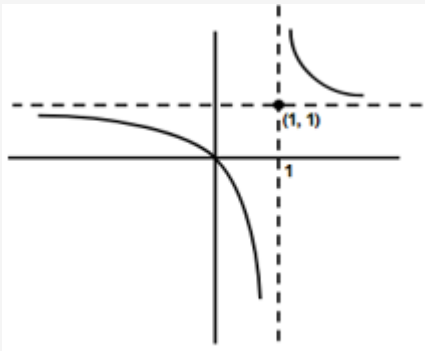
Q80: Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be defined as $f(x) = 2x - 1$ and $g : \mathbb{R} - \{1\} \rightarrow \mathbb{R}$ be defined as $g(x) = \frac{x - \frac{1}{2}}{x - 1}$. Then the composition function $f(g(x))$ is

- (A) One-one but not onto
- (B) Both one-one and onto
- (C) Neither one-one nor onto
- (D) Onto but not one-one

Q80: Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be defined as $f(x) = 2x - 1$ and $g : \mathbb{R} - \{1\} \rightarrow \mathbb{R}$ be defined as $g(x) = \frac{x - \frac{1}{2}}{x - 1}$. Then the composition function $f(g(x))$ is

- (A) One-one but not onto
- (B) Both one-one and onto
- (C) Neither one-one nor onto
- (D) Onto but not one-one

Solution:



$$\begin{aligned} f(g(x)) &= 2g(x) - 1 \\ &= 2 \frac{\left(x - \frac{1}{2}\right)}{x-1} - 1 = \frac{x}{x-1} \\ f(g(x)) &= 1 + \frac{1}{x-1} \end{aligned}$$

One-one, into

Q81: Let three vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$ be such that $\vec{c}$ is coplanar with $\vec{a}$ and $\vec{b}$, $\vec{a} \cdot \vec{c} = 7$ and $\vec{b}$ is perpendicular to $\vec{c}$ where $\vec{a} = -\hat{i} + \hat{j} + \hat{k}$ and $\vec{b} = 2\hat{i} + \hat{k}$, then the value of $2|\vec{a} + \vec{b} + \vec{c}|^2$ is _____.

Q81: Let three vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$ be such that $\vec{c}$ is coplanar with $\vec{a}$ and $\vec{b}$, $\vec{a} \cdot \vec{c} = 7$ and $\vec{b}$ is perpendicular to $\vec{c}$ where $\vec{a} = -\hat{i} + \hat{j} + \hat{k}$ and $\vec{b} = 2\hat{i} + \hat{k}$, then the value of $2|\vec{a} + \vec{b} + \vec{c}|^2$ is _____.

75.00

Solution:

$$\vec{c} = \lambda (\vec{b} \times (\vec{a} \times \vec{b}))$$

$$\vec{c} = \lambda ((\vec{b} \cdot \vec{b}) \vec{a} - (\vec{b} \cdot \vec{a}) \vec{b})$$

$$= \lambda (5(-\hat{i} + \hat{j} + \hat{k}) + 2\hat{i} + \hat{k})$$

$$= \lambda (-3\hat{i} + 5\hat{j} + 6\hat{k})$$

$$\vec{c} \cdot \vec{a} = 7 \Rightarrow 3\lambda + 5\lambda + 6\lambda = 7$$

$$\lambda = \frac{1}{2}$$

$$\therefore 2 \left| \left(\frac{-3}{2} - 1 + 2 \right) \hat{i} + \left(\frac{5}{2} + 1 \right) \hat{j} + (3 + 1 + 1) \hat{k} \right|^2$$

$$= 2 \left(\frac{1}{4} + \frac{49}{4} + 25 \right) = 25 + 50 = 75$$

Q82: If the least and the largest real values of α , for which the equation $z + \alpha |z - 1| + 2i = 0$ ($z \in \mathbb{C}$ and $i = \sqrt{-1}$) has a solution, are p and q respectively; then $4(p^2 + q^2)$ is equal to

Q82: If the least and the largest real values of α , for which the equation $z + \alpha |z - 1| + 2i = 0$ ($z \in C$ and $i = \sqrt{-1}$) has a solution, are p and q respectively; then $4(p^2 + q^2)$ is equal to

10.00

Solution:

$$x + iy + \alpha \sqrt{(x-1)^2 + y^2} + 2i = 0$$

$$\therefore y + 2 = 0 \text{ and } x + \alpha \sqrt{(x-1)^2 + y^2} = 0$$

$$y = -2 \text{ \& } x^2 = \alpha^2 (x^2 - 2x + 1 + 4)$$

$$\alpha^2 = \frac{x^2}{x^2 - 2x + 5}$$

$$\alpha^2 \in \left[0, \frac{5}{4}\right]$$

$$\therefore \alpha^2 \in \left[0, \frac{5}{4}\right]$$

$$\therefore \alpha \in \left[-\frac{\sqrt{5}}{2}, \frac{\sqrt{5}}{2}\right]$$

$$\text{Then } 4(p^2 + q^2) = 4\left(\frac{5}{4} + \frac{5}{4}\right) = 10$$

Q83: If $\int_{-a}^a (|x| + |x - 2|) dx = 22$, ($a > 2$) and $[x]$ denotes the greatest integer $\leq x$,

then $\int_a^{-a} (x + [x]) dx$ is equal to _____.

Q83: If $\int_{-a}^a (|x| + |x - 2|) dx = 22$, ($a > 2$) and $[x]$ denotes the greatest integer $\leq x$, then $\int_a^{-a} (x + [x]) dx$ is equal to _____.

3.00

Solution:

$$\int_{-a}^0 (-2x + 2) dx + \int_0^2 (x + 2 - x) dx + \int_2^a (2x - 2) dx = 22$$

$$x^2 - 2x|_0^{-a} + 2x|_0^2 + x^2 - 2x|_2^a = 22$$

$$a^2 + 2a + 4 + a^2 - 2a - (4 - 4) = 22$$

$$2a^2 = 18 \Rightarrow a = 3$$

$$\int_{-3}^3 (x + [x]) dx = -3 - 2 - 1 + 1 + 2 = -3$$

Q84: Let M be any 3×3 matrix with entries from the set $\{0, 1, 2\}$. The maximum number of such matrices, for which the sum of diagonal elements of $M^T M$ is seven, is _____.

Q84: Let M be any 3×3 matrix with entries from the set $\{0, 1, 2\}$. The maximum number of such matrices, for which the sum of diagonal elements of $M^T M$ is seven, is _____.

540.00

Solution:

$$\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \begin{bmatrix} a & d & g \\ b & e & h \\ c & f & i \end{bmatrix}$$

$$a^2 + b^2 + c^2 + d^2 + e^2 + f^2 + g^2 + h^2 + i^2 = 7$$

Case I: Seven (1's) and two (0's)

$${}^9C_2 = 36$$

Case II: One (2) and three (1's) and five (0's)

$$\frac{9!}{5!3!} = 504$$

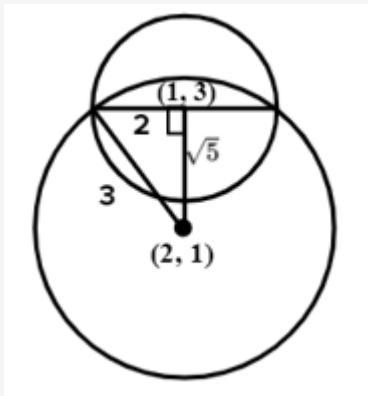
$\therefore$ Total = 540

Q85: If one of the diameters of the circle $x^2 + y^2 - 2x - 6y + 6 = 0$ is a chord of another circle 'C'. whose center is at (2, 1), then its radius is_____.

Q85: If one of the diameters of the circle $x^2 + y^2 - 2x - 6y + 6 = 0$ is a chord of another circle 'C'. whose center is at (2, 1), then its radius is_____.

3.00

Solution:



Distance between (1, 3) and (2, 1) is $\sqrt{5}$

$$\therefore (\sqrt{5})^2 + (2)^2 = r^2$$

$$\Rightarrow r = 3$$

Q86: The minimum value of α for which the equation $\frac{4}{\sin x} + \frac{1}{1-\sin x} = \alpha$ has at least one solution in $(0, \frac{\pi}{2})$ is _____.

Q86: The minimum value of α for which the equation $\frac{4}{\sin x} + \frac{1}{1-\sin x} = \alpha$ has at least one solution in $(0, \frac{\pi}{2})$ is _____.

9.00

Solution:

$$\text{Let } f(x) = \frac{4}{\sin x} + \frac{1}{1-\sin x}$$

$$y = \frac{4-3\sin x}{\sin x(1-\sin x)}$$

$$\text{Let } \sin x = t \text{ when } t \in (0, 1)$$

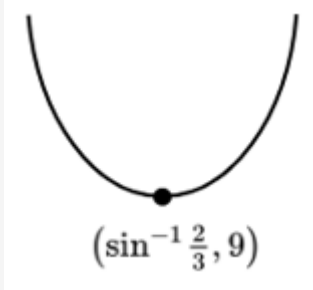
$$y = \frac{4-3t}{t-t^2}$$

$$\frac{dy}{dt} = \frac{-3(t-t^2)-(1-2t)(4-3t)}{(t-t^2)^2} = 0$$

$$\Rightarrow 3t^2 - 3t - (4 - 11t + 6t^2) = 0$$

$$\Rightarrow 3t^2 - 8t + 4 = 0$$

$$\Rightarrow 3t^2 - 6t - 2t + 4 = 0$$



$$\Rightarrow t = \frac{2}{3}$$

$$\Rightarrow \alpha \geq 9$$

Least α is equal to 9

Q87: Let $A = \{n \in N : n \text{ is a } 3\text{-digit number}\}$

$B = \{9k + 2 : k \in N\}$ and

$C = \{9k + l : k \in N\}$ for some l ($0 < l < 9$)

If the sum of all the elements of the set $A \cap (B \cup C)$ is 274×400 , then l is equal to _____

Q87: Let $A = \{n \in N : n \text{ is a } 3\text{-digit number}\}$

$B = \{9k + 2 : k \in N\}$ and

$C = \{9k + l : k \in N\}$ for some l ($0 < l < 9$)

If the sum of all the elements of the set $A \cap (B \cup C)$ is 274×400 , then l is equal to _____

Solution:

3 digit number of the form $9k + 2$ are $\{101, 109, \dots, 992\}$

$\Rightarrow$ Sum equal to $\frac{100}{2}(1093)$

Similarly sum of 3 digit number of the form $9k + 5$ is $\frac{100}{2}(1099)$

$$\frac{100}{2}(1093) + \frac{100}{2}(1099) = 100 \times (1096)$$

$$= 400 \times 274$$

$$\Rightarrow l = 5$$

Q88: Let B_i ($i = 1, 2, 3$) be three independent events in a sample space. The probability that only B_1 occurs is α , only B_2 occurs is β and only B_3 occurs is γ . Let p be the probability that none of the events B_i occurs and these 4 probabilities satisfy the equations $(\alpha - 2\beta)p = \alpha\beta$ and $(\beta - 3\gamma)p = 2\beta\gamma$ (All the probabilities are assumed to lie in the interval $(0, 1)$). Then $\frac{P(B_1)}{P(B_3)}$ is equal to _____.

Q88: Let B_i ($i = 1, 2, 3$) be three independent events in a sample space. The probability that only B_1 occurs is α , only B_2 occurs is β and only B_3 occurs is γ . Let p be the probability that none of the events B_i occurs and these 4 probabilities satisfy the equations $(\alpha - 2\beta)p = \alpha\beta$ and $(\beta - 3\gamma)p = 2\beta\gamma$ (All the probabilities are assumed to lie in the interval $(0, 1)$). Then $\frac{P(B_1)}{P(B_3)}$ is equal to _____.

Solution:

Let x, y, z be probability of B_1, B_2, B_3 respectively

$$\Rightarrow x(1-y)(1-z) = \alpha \Rightarrow y(1-x)(1-z) = \beta$$

$$\Rightarrow z(1-x)(1-y) = \gamma \Rightarrow (1-x)(1-y)(1-z) = p$$

Putting in the given relation we get $x = 2y$ and $y = 3z \Rightarrow x = 6z \Rightarrow \frac{x}{z} = 6$

Q89: $\lim_{n \rightarrow \infty} \tan \left\{ \sum_{r=1}^n \tan^{-1} \left(\frac{1}{1+r+r^2} \right) \right\}$ is equal to _____.

Q89: $\lim_{n \rightarrow \infty} \tan \left\{ \sum_{r=1}^n \tan^{-1} \left(\frac{1}{1+r+r^2} \right) \right\}$ is equal to _____.

1.00

Solution:

$$\begin{aligned} & \lim_{n \rightarrow \infty} \tan \left(\sum_{r=1}^n \tan^{-1} \left(\frac{1}{1+r+r^2} \right) \right) \\ &= \lim_{n \rightarrow \infty} \tan \left(\tan^{-1} (n+1) - \tan^{-1} (1) \right) \\ &= \lim_{n \rightarrow \infty} \tan \left(\tan^{-1} \left(\frac{n}{n+2} \right) \right) \\ &= \tan \left(\tan^{-1} (1) \right) \\ &= \tan \frac{\pi}{4} = 1 \end{aligned}$$

Q90: Let $P = \begin{bmatrix} 3 & -1 & -2 \\ 2 & 0 & \alpha \\ 3 & -5 & 0 \end{bmatrix}$, where $\alpha \in \mathcal{R}$. Suppose $Q = [q_{ij}]$ is a matrix satisfying $PQ = kI_3$ for some non-zero $k \in \mathcal{R}$. If $q_{23} = -\frac{k}{8}$ and $|Q| = \frac{k^2}{2}$, then $\alpha^2 + k^2$ is equal to

_____.

Q90: Let $P = \begin{bmatrix} 3 & -1 & -2 \\ 2 & 0 & \alpha \\ 3 & -5 & 0 \end{bmatrix}$, where $\alpha \in \mathbb{R}$. Suppose $Q = [q_{ij}]$ is a matrix satisfying $PQ = kI_3$ for some non-zero $k \in \mathbb{R}$. If $q_{23} = -\frac{k}{8}$ and $|Q| = \frac{k^2}{2}$, then $\alpha^2 + k^2$ is equal to

17.00

Solution:

$$\text{As } PQ = kI \Rightarrow Q = kP^{-1}I$$

$$\text{Now } Q = \frac{k}{|P|} (\text{adj } P) I \Rightarrow Q = \frac{k}{(20+12\alpha)} \begin{bmatrix} - & - & - \\ - & - & (-3\alpha-4) \\ - & - & - \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\therefore q_{23} = \frac{-k}{8} \Rightarrow \frac{k}{(20+12\alpha)}(-3\alpha-4) = \frac{-k}{8}$$

$$\Rightarrow 2(3\alpha+4) = 5+3\alpha$$

$$3\alpha = -3 \Rightarrow \alpha = -1$$

$$\text{Also } |Q| = \frac{k^3 |I|}{|P|} \Rightarrow \frac{k^2}{2} = \frac{k^3}{(20+12\alpha)}$$

$$(20+12\alpha) = 2k \Rightarrow 8 = 2k \Rightarrow k = 4$$

